

Radiology Voice-to-Structured-Report System

Technical Design Document — V1 through Full-Scale Release

Version: 1.0 — pet-project status noted; Model Config Admin Panel added **Scope:** Single hospital / group deployment **Status:** Pre-implementation

Changelog v0.9 → v1.0. Project status noted as a personal/pet project for now (§15.0a) — commercial offering to labs is contingent on achieving real cost optimization, which this version’s new feature exists to enable. Added the **Model Config Admin Panel** (§7.2 component, §7.9.6 behaviour, §6.14 data model): a per-task, per-model configuration screen covering both cloud and local open-source models uniformly, gated by mandatory gold-set testing before any swap goes live.

Changelog v0.8 → v0.9. Correction, not a refinement. The current transcriptionist cost baseline used throughout §15 was ₹40–60/report; the client has now confirmed the real figure is **₹12/report**. This inverts the earlier conclusion: the AI system (₹18.5–30.5/report depending on review level) costs *more* than the process it would replace, not less. §15 rewritten to retract the cost-savings pitch and state plainly that pricing now depends entirely on the radiologist-time-saved measurement (§15.2), which was previously the secondary, optional story. Pending: confirming whether ₹12 is wage-only or fully loaded — see §15.1a.

Changelog v0.7 → v0.8. Transcriptionist review cost revised from an estimated ₹30 (full review) / ₹11 (confidence-weighted blend) to a **measured ₹12/report flat**, applied throughout §11.3 and §15.4. This replaces an assumption with a real client-provided number — treat it accordingly if it later turns out to vary by report complexity.

Changelog v0.6 → v0.7. Prior-study comparison deferred to customer request, not GA scope (D9 — accession numbers don’t chain across a patient’s studies). Autonomy-class grouping clarified as a cost-timeline lever, not an accuracy lever (D11). Baseline audit (D12) folded into the onboarding flow as a formal step. Quality-audit sampling schedule specified in full, grading reassigned to the transcriptionist (D13). Tenant isolation upgraded from “column now, enforcement later” to **real enforcement from day one** (D19) — application-layer scoping plus row-level security, with schema-level per-tenant separation planned as a later, client-managed migration.

Changelog v0.5 → v0.6. Resolved D9–D19 (§14). Key changes: templates confirmed as Word/PDF imports (D16); historical corpus confirmed to carry no template mapping, S2 derivation path confirmed necessary (D17); baseline CSE audit method specified with a worked example (D12); and a genuinely new decision — **plan is single hospital now, multi-tenant (100+) later (D19)**. Added lightweight `tenant_id` scoping to core tables now, deferring full multi-tenancy build (billing, admin, isolation hardening) until a second client is signed. See §6.13 and §7.10.

Changelog v0.4 → v0.5. Added §7.9 (Model Selection: Frontier vs. Open-Source) covering the full comparison across Claude, GPT, Gemini, and Kimi tiers; the recommended per-task model split; and open-source local-hosting options for the radiologist-workstation tasks, including a correction that Meta’s Muse Spark is proprietary, not open-source, despite naming that suggests otherwise. §11 Cost Model rebuilt around the recommended split with a volume table.

Changelog v0.3 → v0.4. Added §8.6 (training data lifecycle and adaptation roadmap) and §6.12 (adaptation tables). Makes explicit that **no model is trained at onboarding** and why. Adds six prerequisite gates before any ASR fine-tune. Fixed a stale §6.10 cross-reference introduced in v0.3.

Changelog v0.2 → v0.3. Added the Knowledge Onboarding Console (§7.8, §6.10) — corpus, template, shorthand, and paired-audio import promoted from scattered engineering scripts to a versioned, re-runnable product module with clinical approval gates. Added `spoken_study_code` to `template_version` (missing from the D5 change). Weeks 1–5 now execute *through* the console rather than around it.

Changelog v0.1 → v0.2. DICOM deferred behind a measured trigger (§4.4). Spoken study-code phrase adopted as the primary routing anchor (§8.3.9). Microphone standardisation moved to week 1 and the gold set split by capture hardware (§5.3). `absence_policy` split into a two-pass process (§8.3.6). Guideline modules removed from scope. Autonomy redesigned around statistical non-inferiority at the *class* level with automatic revocation (§8.3.10). Audio retained indefinitely as lossless FLAC under a dual legal basis (§6.10). Critical-findings detection promoted from gap to V1 scope.

1. Purpose

Convert a radiologist’s unstructured spoken findings (Indian-accented English, recorded and uploaded) into a completed structured report conforming to one of 50–200 predefined templates, at an accuracy level that does not regress against the existing human-transcriptionist process.

2. Context and Constraints

Dimension	Value	Design consequence
Data residency	Cloud APIs permitted	Frontier ASR/LLM available; still minimise PHI egress
Capture mode	Recorded, then uploaded	Fully batch; latency budget is generous (60–180 s/report)
Unit of work	1 recording = 1 report	No session segmentation problem

Metadata available	Patient name, age, gender, referring doctor	No modality/body part/protocol — must be inferred or sourced from DICOM
DICOM access	Not yet integrated (recommended)	Highest-ROI integration available
Templates	50-200, fixed schema each	Routing is a real classification problem
Code words	Domain shorthand (UGS, LMC, ...)	Lexicon/expansion layer, not control tokens
Labelled data	~few hundred audio + signed report pairs	Not enough for ASR fine-tune yet; enough for eval set
Unlabelled data	Historical signed reports (thousands)	Lexicon mining, exemplars, synthetic transcripts
Incumbent	Human transcriptionist (~2-3% clinically significant error)	Strong baseline — AI must augment, not replace, on day one
Environment	Reading room, some noise	Microphone standardisation is a first-class work item
Optimisation target	Accuracy over latency	Ensembles, self-consistency, verification passes are affordable
Capture hardware	Standardised dictation mic, budget approved	Moves to week 1; gold set must be split by hardware generation
Study-code phrase	Adopted — spoken at start of every dictation	Primary routing anchor; largely substitutes for DICOM
Guideline modules	Out of scope	Keeps CDSCO classification in the low-risk band
Audio retention	Indefinite, lossless	Fine-tuning corpus; requires a separate legal basis from clinical retention

3. Design Invariants

These are non-negotiable and every component below is shaped by them.

- I1 — Provenance or nothing.** Every clinical assertion in the generated report must carry a pointer to a character span in the transcript and a time span in the audio. Any assertion the pipeline cannot ground is discarded and flagged, never printed.
- I2 — Classify, never delete.** No stage destructively removes content. Utterances are labelled and filtered downstream. The full transcript is always retained for audit and recovery.
- I3 — Deterministic where possible.** The orchestration graph is a fixed state machine. LLMs occupy specific, bounded nodes. Exactly one cycle exists in the graph (verifier → targeted re-extraction), capped at 2 iterations.
- I4 — Human in the loop, always at v1.** The system produces drafts. Autonomy is granted per template class, only after measured evidence, and can be revoked per class.
- I5 — Everything is versioned.** Templates, prompts, model identifiers, lexicons, and schemas all carry versions. Every report records the exact versions that produced it.
- I6 — The eval harness precedes the pipeline.** No component ships without a measurement of its effect on the frozen evaluation set.

4. Release Plan

4.1 V1 — Pilot (Weeks 1-8)

Product shape: Audio → AI draft → transcriptionist edits → radiologist signs

The AI is inserted *upstream* of the existing transcriptionist, not in place of her. This makes safety regression structurally impossible, produces immediate ROI in transcription minutes, and starts the correction flywheel from day one.

In scope - Knowledge Onboarding Console (\$7.8) — templates, corpus, shorthand, paired audio, rosters, boilerplate, critical rules - Standardised push-to-talk dictation microphones deployed (week 1) - Ingest API + object storage (lossless FLAC) + durable job queue - VAD, single ASR engine (benchmarked choice) with keyterm biasing - **Study-code phrase detection** — the primary routing anchor - Code word / abbreviation expansion layer - Study-code → lexical → demographic routing cascade (top ~20 templates only) - Template-bound extraction with provenance enforcement - Schema validation + rule-based validators - **Critical-findings detection and alerting** (\$9.1 — promoted to V1) - Transcriptionist review UI with per-field audio playback - Edit capture (the flywheel) - Eval harness with frozen gold set, split by capture hardware

Out of scope for V1 Multi-engine ensemble, phonetic post-correction, LLM verifier, DICOM, per-speaker adaptation, autonomy, RIS export, prior comparison.

Why critical findings moved into V1. Introducing a draft queue creates a delay between the radiologist speaking a red-flag finding and anyone acting on it — an exposure the current transcriptionist process does not have. The risk is created the moment reports start queueing, so the mitigation ships with the queue.

Exit criteria | Metric | Target | |—|—| | Transcriptionist minutes per report | ≥30% reduction vs. baseline | | Reports where draft was discarded entirely | <10% | | Template routing top-1 accuracy (pilot templates) | ≥95% | | Schema-valid output | 100% | | Clinically significant errors reaching signature | No increase vs. baseline | | CODEWORD_COMPLIANCE (study code spoken) | **Measured from day one — decides D1** | | STUDYCODE_RECALL (spoken → detected) | ≥98% | | Critical-finding alert recall on gold set | 100% | | Edit pairs captured | ≥1,500 |

4.2 Beta (Months 3-6)

Product shape: same, plus a narrow autonomy pilot on one or two high-confidence template classes.

Added - DICOM integration — conditional, see §4.4 - Multi-engine ASR fan-out + reconciliation - Phonetic post-correction against mined lexicon - Utterance segmentation, classification, and self-correction repair resolution - LLM verifier + round-trip semantic check - Confidence aggregation and field-level flagging - Per-speaker profiles (term lists, exemplars, error patterns) - Full template library coverage - Autonomy accrual begins (observation only — no grants yet) - Shadow-mode infrastructure for safe version rollout

Exit criteria | Metric | Target | |—|—| | Transcriptionist minutes per report | ≥55% reduction | | Clinically significant error rate in draft | ≤5% | | Verifier recall on injected clinically significant errors | ≥90% | | Routing top-1 (full library) | ≥99% | | Autonomy accrual | ≥1 class reaching accruing with clean interim posterior | | Edit pairs captured | ≥15,000 |

4.3 Full Scale / GA (Months 6-12+)

Product shape: graduated autonomy. Template classes move to AI → radiologist as their measured error rate earns it, one class at a time, each revocable.

Added - ASR adaptation on captured verbatim corrections, per-speaker where volume allows — **only once the §8.6.5 prerequisite gates pass** - Distilled small models for utterance classification and routing - HL7 v2 ORU^R01 / FHIR DiagnosticReport export to RIS - ~~Prior-study retrieval and comparison section generation~~ — **deferred, build only if a customer requests it (D9)**: accession numbers are per-report with no chaining across a patient’s studies, so this needs a separate patient-identity linkage that doesn’t exist yet - Critical-findings detection and alerting workflow - Amendment/addendum workflow - Drift monitoring and automated regression gates - Autonomy grant/revocation engine (§8.3.10)

Guideline modules (Fleischner, TI-RADS, LI-RADS) are explicitly out of scope. They convert the product from transcription assistance into clinical decision support and change its regulatory classification. Revisit only as a separate product decision with its own legal review.

Exit criteria | Metric | Target | |—|—| | Reports on autonomous path | ≥40% by volume | | Clinically significant errors reaching signature | Non-inferior at 1 pp margin vs. measured baseline | | End-to-end turnaround | <10 min p95 | | Transcriptionist effort | Reallocated to complex/low-confidence tail only |

4.4 The DICOM decision (D1)

Adopting a spoken study code absorbs most of DICOM’s value, because both are deterministic routing anchors. Expected marginal contribution:

Signals available	Expected ROUTE_TOP1
Lexical fingerprint + demographics only	95-98%
+ study-code phrase (when spoken and recognised)	~99.5%
+ DICOM	~99.9%

DICOM’s marginal routing gain over the study code is roughly half a percentage point — at 200 studies/day, one wrong-template report every week or two. Not worth an integration project on its own.

What actually decides it is compliance, not recognition. ASR will detect the code reliably; the radiologist forgetting to say it is the larger hole, and new verbal habits typically settle at 80-95% in the first months. DICOM never forgets.

Trigger rule. Build V1 without DICOM. Instrument CODEWORD_COMPLIANCE from pilot day one. After four weeks of pilot: **≥95% → defer DICOM indefinitely.** **<90% → pursue in Beta.** Between the two, re-measure after a compliance-coaching cycle.

Separate dependency, unrelated to accuracy. Upload metadata currently carries no accession number. Without one, reports cannot be filed back to the RIS at GA. Confirm accession availability in week 1 — if it is absent, DICOM or a RIS query returns to the roadmap for integration reasons regardless of what routing accuracy says.

5. Metrics and Evaluation

5.1 Why WER is not the target

WER weights “the” vs “a” identically to “no” vs “a”. The metrics below are the ones that govern release decisions.

5.2 Metric definitions

Metric	Definition	Owner stage
WER	Standard word error rate vs. verbatim gold transcript	ASR
INS_RATE	Insertion rate specifically — the hallucination signal	ASR
CTER	Clinical Term Error Rate: errors restricted to anatomy, pathology, laterality, numerals, negation cues	ASR
CODEWORD_RECALL	Code words spoken and correctly detected ÷ spoken	Normalisation
CODEWORD_COMPLIANCE	Recordings where a study code was spoken at all ÷ total. Per radiologist. Decides D1	Behavioural
STUDYCODE_RECALL	Study codes spoken and correctly detected ÷ spoken	Normalisation
CF_RECALL	Critical findings present and alerted ÷ present. Target 100%	Critical findings
CF_SLA_BREACH	Alerts unacknowledged past <code>sla_minutes</code> ÷ alerts	Critical findings
UTT_F1	Per-class F1 for utterance labels (report / correction / command / aside / other-speaker)	Segmentation
REPAIR_ACC	Self-corrections correctly resolved to the intended value	Repair
ROUTE_TOP1 / ROUTE_TOP3	Template routing accuracy, reported per template , never averaged only	Routing
FIELD_ACC	Per-field exact/semantic match vs. gold structured report	Extraction
FIELD_F1	Macro F1 across template fields	Extraction
HALLUC_RATE	Assertions in output not entailed by transcript ÷ total assertions	Composition
ORPHAN_RATE	Clinical assertions in transcript with no landing field	Coverage
CSE_DRAFT	Reports containing ≥1 clinically significant error at draft handoff	End-to-end
CSE_SIGNED	Reports containing ≥1 clinically significant error at signature	Primary safety metric
EDIT_MINUTES	Median transcriptionist minutes per report	Primary value metric
EDIT_DISTANCE	Character-level edit distance draft → final	End-to-end
VERIFIER_RECALL	Injected clinically significant errors caught by verifier	Verification

5.3 Gold evaluation set

Split by capture hardware. The existing few hundred recordings were captured on the old setup. Building the frozen gold set from them alone would rank ASR engines on acoustic conditions you are about to stop having, and engine rankings genuinely shift between noisy and clean audio.

Partition	Size	Source	Used for
legacy	100	Existing archive recordings	Evaluating reprocessing of the historical archive
current	150	Recorded on the new standardised microphones	All forward-looking decisions , including the week-3 ASR bake-off

This makes microphone procurement a **week-1 dependency**, ahead of the bake-off rather than parallel to it.

- **Frozen.** Never trained on. Refreshed additively every 6 months, retaining original items so trends stay comparable.
- **Stratification:** modality × template × speaker × audio quality bucket.
- **Annotation layers per item:** verbatim transcript (disfluencies, false starts, corrections, asides all retained), utterance labels, code word occurrences, gold template ID, gold field values, clinically-significant-error markings.
- **Source of annotation:** your existing transcriptionists, briefed that *verbatim* differs from their normal cleaned output.
- **Dev set:** remaining paired recordings, plus synthetic transcripts generated from historical signed reports.

5.4 Clinical significance rubric

Errors are graded by a human, not by the system, using a fixed rubric. **Grading is done by the transcriptionist (D13)**, not the radiologist — she is already reading every report, so this uses existing effort rather than adding a new reviewer role.

Grade	Definition
G0	Cosmetic — grammar, formatting, style
G1	Non-clinical content error — no change to meaning
G2	Clinical content error, low impact — would not alter management

G3	Clinical content error, would alter management
G4	Reversal of meaning, wrong laterality, wrong patient context, fabricated finding

5.4.1 SAMPLING SCHEDULE (D13)

Grading every report indefinitely isn’t necessary once the system is trusted; the check-rate steps down over the first several months as evidence accumulates. Applied per client, starting from that client’s pilot go-live:

Period	Duration	Sampling rate
Weeks 1-4	4 weeks	Every report
Weeks 5-12	6-8 weeks	1 in 5
Weeks 13-16	4 weeks	1 in 10
Months 5-7	3 months	1 in 20

Beyond month 7, continue at 1-in-20 unless `CUSUM` monitoring (§8.3.10) or a spike in `edit_distance` flags a class for renewed full-rate checking. The schedule resets to week-1 rates for any `autonomy_class` newly granted autonomy, since a class’s transition off human review is exactly the moment tighter checking matters most.

`CSE_*` metrics count `G3 + G4`.

6. Data Model

PostgreSQL 16 + `pgvector`. Conventions: `snake_case`, surrogate `uuid` primary keys, `created_at` / `updated_at` on every table, soft delete via `deleted_at` where retention matters.

6.1 Entity overview



6.2 Identity and clinical context

`APP_USER`

System users. Roles are additive.

Column	Type	Key / Constraint	Notes
<code>id</code>	<code>uuid</code>	PK	
<code>employee_code</code>	<code>text</code>	UNIQUE NOT NULL	Hospital HR identifier
<code>display_name</code>	<code>text</code>	NOT NULL	
<code>email</code>	<code>citext</code>	UNIQUE	
<code>roles</code>	<code>text[]</code>	NOT NULL	{radiologist, transcriptionist, admin, auditor}
<code>is_active</code>	<code>boolean</code>	NOT NULL DEFAULT true	
<code>created_at</code>	<code>timestampz</code>	NOT NULL	

`RADIOLOGIST_PROFILE`

Speech-specific configuration. One row per dictating radiologist.

Column	Type	Key / Constraint	Notes
--------	------	------------------	-------

id	uuid	PK	
user_id	uuid	FK → app_user.id , UNIQUE	
voice_embedding	vector(192)	NULL	Speaker enrollment for diarization. Biometric data — see §10
voice_enrolled_at	timestampz	NULL	
voice_consent_ref	text	NULL	Consent artefact reference
default_language	text	NOT NULL DEFAULT 'en-IN'	
subspecialty	text[]		Routing prior
autonomy_enabled	boolean	NOT NULL DEFAULT false	Per-radiologist autonomy gate
asr_adapter_ref	text	NULL	Per-speaker fine-tuned adapter (GA)

PATIENT

Minimal PHI. Stored encrypted at column level; never sent to third-party APIs.

Column	Type	Key / Constraint	Notes
id	uuid	PK	
mrn	text	UNIQUE NOT NULL, encrypted	Hospital medical record number
name_enc	bytea	encrypted	Never leaves the VPC
pseudonym	text	UNIQUE NOT NULL	Stable token used in all LLM prompts
date_of_birth	date	NULL	
age_years	smallint		Denormalised for routing
sex	text	CHECK IN (M , F , O , U)	Routing filter + validator
created_at	timestampz	NOT NULL	

STUDY

The imaging examination. Populated from upload metadata and enriched from DICOM when available.

Column	Type	Key / Constraint	Notes
id	uuid	PK	
patient_id	uuid	FK → patient.id	
accession_number	text	UNIQUE	RIS linkage key. Confirmed per-report, not chained across a patient's studies (D9) — do not use this field to attempt prior-study linkage; see §4.3
study_instance_uid	text	UNIQUE NULL	DICOM (0020,000D)
study_datetime	timestampz		
referring_doctor	text		Routing prior
modality	text	NULL	DICOM (0008,0060) — coded, reliable
body_part_examined	text	NULL	DICOM (0018,0015) — often blank
study_description	text	NULL	DICOM (0008,1030) — free text but stable per site
protocol_name	text	NULL	DICOM (0018,1030)
contrast_agent	text	NULL	DICOM (0018,0010) — validator input
series_count	smallint	NULL	
metadata_source	text	CHECK IN (upload , dicom , inferred)	Provenance of routing inputs
priority	text	CHECK IN (routine , urgent , stat) DEFAULT routine	Queue ordering
prior_study_ids	uuid[]		Deferred (D9) — comparison-section source, on hold pending a customer request and a real patient-identity linkage strategy

Index: (patient_id, study_datetime DESC) for prior retrieval; (accession_number) ; (study_description) for the routing lookup table.

6.3 Ingestion

RECORDING

One audio file. One report.

Column	Type	Key / Constraint	Notes
id	uuid	PK	
study_id	uuid	FK → study.id , UNIQUE	Enforces 1:1
radiologist_id	uuid	FK → radiologist_profile.id	Declared speaker
object_key	text	NOT NULL	Object storage path
content_hash	text	UNIQUE NOT NULL	Idempotency key — rejects duplicate uploads
duration_seconds	numeric(8,2)		
sample_rate_hz	integer		Quality gate: warn if <16000
channels	smallint		
codec	text		Quality gate: warn on lossy
bitrate_kbps	integer		
measured_snr_db	numeric(5,2)	NULL	Computed at ingest; correlates with error rate
device_id	text	NULL	Microphone tracking
capture_device_class	text	CHECK IN (legacy , dictation_mic_ptt , head set)	Gold-set partitioning; stratifies every metric
is_push_to_talk	boolean	NOT NULL DEFAULT false	PTT eliminates most pre-roll/aside contamination at source
audio_format	text	NOT NULL	Must be flac or wav . Lossy formats rejected at ingest
study_code_spoken_text	text	NULL	Raw detected carrier phrase
study_code_template_id	uuid	FK → template.id NULL	Deterministic routing anchor
study_code_confidence	numeric(5,4)	NULL	
end_marker_detected	boolean	NOT NULL DEFAULT false	Truncation detection + post-roll trim
uploaded_at	timestamptz	NOT NULL	
is_training_corpus_eligible	boolean	NOT NULL DEFAULT false	Separate legal basis from clinical retention — see §6.11
retention_expires_at	timestamptz	NULL	NULL = indefinite (clinical archive)

Note: content_hash as a unique constraint is the cheapest possible protection against double-processing, which will otherwise happen the first time someone retries a failed upload.

6.4 ASR and transcript

ASR_RUN

One execution of one engine against one recording. Multiple rows per recording in Beta+.

Column	Type	Key / Constraint	Notes
id	uuid	PK	
recording_id	uuid	FK → recording.id	
engine	text	NOT NULL	deepgram_nova3_medical , azure_custom_v4 , whisper_medium_ft_v3
engine_version	text	NOT NULL	Pinned. Never “latest”.
config_hash	text	NOT NULL	Hash of keyterm list + params, for reproducibility
keyterm_set_id	uuid	FK → lexicon_set.id	Which bias list was applied
raw_response	jsonb		Full vendor payload, retained
overall_confidence	numeric(5,4)		
wer_estimate	numeric(5,4)	NULL	Populated only for eval items
latency_ms	integer		
status	text	CHECK IN (pending , ok , failed)	
created_at	timestamptz	NOT NULL	

UNIQUE (recording_id, engine, engine_version, config_hash) — makes reruns idempotent.

ASR_SEGMENT

Word- or phrase-level output with timing and confidence. Partitioned by month.

Column	Type	Key / Constraint	Notes
id	bigserial	PK	
asr_run_id	uuid	FK → asr_run.id	
seq	integer	NOT NULL	Order within run
text	text	NOT NULL	
start_ms	integer	NOT NULL	
end_ms	integer	NOT NULL	
confidence	numeric(5,4)		Per-token confidence
speaker_label	text	NULL	Diarization output
is_primary_speaker	boolean	NULL	Resolved against voice enrollment
alternatives	jsonb	NULL	N-best for this position

Index: (asr_run_id, seq) . Consider jsonb consolidation on asr_run instead if row counts become a problem; keep the relational form while the reconciliation algorithm is still being tuned.

TRANSCRIPT

The reconciled, normalised transcript. One canonical row per recording, versioned.

Column	Type	Key / Constraint	Notes
id	uuid	PK	
recording_id	uuid	FK → recording.id	
version	integer	NOT NULL	Increments on reprocessing
stage	text	CHECK IN (raw , reconciled , normalised , re paired)	Snapshot at each transform
text	text	NOT NULL	Full text; char offsets referenced by provenance
source_asr_run_ids	uuid[]	NOT NULL	Which engines contributed
reconciliation_method	text		single , rover , llm_arbitrated
disagreement_score	numeric(5,4)		Engine disagreement — a strong uncertainty signal
is_current	boolean	NOT NULL	
created_at	timestampz	NOT NULL	

UNIQUE (recording_id, version, stage) . Partial index on is_current = true .

TRANSCRIPT_UTTERANCE

Segmentation + classification. **Nothing is deleted; everything is labelled.** (Invariant I2)

Column	Type	Key / Constraint	Notes
id	uuid	PK	
transcript_id	uuid	FK → transcript.id	
seq	integer	NOT NULL	
char_start	integer	NOT NULL	Offset into transcript.text
char_end	integer	NOT NULL	
audio_start_ms	integer	NOT NULL	
audio_end_ms	integer	NOT NULL	
text	text	NOT NULL	
label	text	NOT NULL	report_content , self_correction , command , aside , disfluency , other_speaker , uncertain
label_confidence	numeric(5,4)	NOT NULL	

label_source	text	CHECK IN (llm , distilled_model , rule , human)	
contains_clinical_tokens	boolean	NOT NULL	Guardrail: non-report label + clinical tokens ⇒ escalate
superseded_by_id	uuid	FK → transcript_utterance.id NULL	Set on the retracted half of a self-correction
is_included_downstream	boolean	NOT NULL	Computed from label + guardrails

Index: (transcript_id, seq) ; partial index on is_included_downstream = true .

6.5 Knowledge: lexicon, code words, templates

LEXICON_SET

A versioned bundle of terms used as ASR bias and correction targets.

Column	Type	Key / Constraint	Notes
id	uuid	PK	
name	text	NOT NULL	global , modality:US , speaker: <uuid>
version	integer	NOT NULL	
scope	text	CHECK IN (global , modality , template , speaker)	
scope_ref	text	NULL	
is_active	boolean	NOT NULL	

LEXICON_TERM

Canonical domain vocabulary. Mined from historical reports, then curated.

Column	Type	Key / Constraint	Notes
id	uuid	PK	
lexicon_set_id	uuid	FK → lexicon_set.id	
canonical_form	text	NOT NULL	last menstrual cycle
short_form	text	NULL	LMC
term_type	text	CHECK IN (anatomy , pathology , abbreviation , code_word , measurement_unit , device , drug , person)	
phonetic_key_primary	text	NOT NULL	Double-metaphone primary
phonetic_key_secondary	text	NULL	Double-metaphone alternate
embedding	vector(768)	NULL	Semantic retrieval
frequency_rank	integer		From historical corpus — biasing priority
radlex_id	text	NULL	Interoperability
snomed_ct_id	text	NULL	
is_ambiguous	boolean	NOT NULL DEFAULT false	PA , RA , CA — needs context resolution
expansion_policy	text	CHECK IN (always_expand , never_expand , per_template)	Does the report say LMC or the full phrase?

Index: GIN on phonetic_key_primary ; ivfflat on embedding . **Critical operational check:** a scheduled job computes pairwise phonetic distance across all term_type = 'code_word' rows and raises an alert on near-collisions. Spelled-out letters cluster into the confusable E-set (B/C/D/E/G/P/T/V/Z), which is exactly where LMC and LMP sit.

LEXICON_SURFACE_VARIANT

How each term actually comes back from ASR. **Mined from real transcripts, not guessed.**

Column	Type	Key / Constraint	Notes
id	uuid	PK	

lexicon_term_id	uuid	FK → lexicon_term.id	
surface_text	text	NOT NULL	you gee ess , U G S , yugs
phonetic_key	text	NOT NULL	
observed_count	integer	NOT NULL DEFAULT 0	Incremented from production
source	text	CHECK IN (mined , manual , generated)	
min_confidence	numeric(5,4)		Threshold below which a match escalates

TEMPLATE / TEMPLATE_VERSION / TEMPLATE_FIELD

template

Column	Type	Key / Constraint	Notes
id	uuid	PK	
code	text	UNIQUE NOT NULL	Stable business key, e.g. US_ABD_ROUTINE
display_name	text	NOT NULL	
modality	text	NOT NULL	Hard routing filter
body_region	text	NOT NULL	
is_module	boolean	NOT NULL DEFAULT false	Composable sub-template (TI-RADS, Lung-RADS)
parent_compatible_codes	text []		Which base templates a module may attach to
applicable_sex	text []		Demographic filter
applicable_age_min / _max	smallint		Demographic filter
usage_count_12m	integer		Power-law head detection — populate this first
autonomy_class_id	uuid	FK → autonomy_class.id NULL	Autonomy is granted per class, not per template — see §8.3.10
is_active	boolean	NOT NULL	

Why not autonomy_tier **per template**. Non-inferiority testing against a ~2.5% baseline needs roughly 3,000 reports per unit of evidence at a 1 pp margin. A template at 1% of volume would take a year to accrue that. Evidence must pool across templates that fail in similar ways, so the grant unit is the class.

template_version

Column	Type	Key / Constraint	Notes
id	uuid	PK	
template_id	uuid	FK → template.id	
version	integer	NOT NULL	
json_schema	jsonb	NOT NULL	Drives constrained decoding
render_spec	jsonb	NOT NULL	Field order, headings, house style
routing_card	text	NOT NULL	Human-written description used for retrieval
routing_card_embedding	vector(768)		
trigger_rules	jsonb	NOT NULL	Metadata + finding conditions
study_description_patterns	text []		DICOM lookup table (if D1 is ever pursued)
spoken_study_code	text	UNIQUE NOT NULL	The phrase the radiologist says: “ultrasound abdomen routine”
spoken_study_code_phonetic	text	NOT NULL	Double-metaphone key; collision-audited across the full set
spoken_study_code_variants	text []		Observed ASR renderings, mined from production
approved_by	uuid	FK → app_user.id	Governance
approved_at	timestamptz		
effective_from	timestamptz	NOT NULL	
is_current	boolean	NOT NULL	

UNIQUE (template_id, version) . Reports are legal records; an immutable version history is mandatory.

template_field

Column	Type	Key / Constraint	Notes
id	uuid	PK	
template_version_id	uuid	FK → template_version.id	
field_key	text	NOT NULL	liver.echotexture
section	text	NOT NULL	Parallel extraction unit
display_label	text	NOT NULL	
data_type	text	CHECK IN (text , enum , measurement , boolean_tri , list)	boolean_tri = present/absent/ uncertain
enum_values	text []	NULL	
unit	text	NULL	
is_required	boolean	NOT NULL	
absence_policy	text	CHECK IN (default_normal , leave_blank_flag , block)	What happens when unmentioned
default_normal_text	text	NULL	
blanket_normal_eligible	boolean	NOT NULL DEFAULT false	May “rest is unremarkable” fill this?
is_critical	boolean	NOT NULL DEFAULT false	Never auto-filled; always human-confirmed
seq	integer	NOT NULL	

absence_policy is a clinical safety decision, not an engineering one. Silently auto-filling “unremarkable” for an organ never mentioned is the highest-liability behaviour in this entire system. Default to leave_blank_flag ; require a radiologist sign-off to move any field to default_normal .

SPEAKER_TERM_BIAS

Per-radiologist vocabulary and error patterns. Populated from the flywheel.

Column	Type	Key / Constraint	Notes
id	uuid	PK	
radiologist_id	uuid	FK → radiologist_profile.id	
lexicon_term_id	uuid	FK → lexicon_term.id	
boost_weight	numeric (4,2)	NOT NULL	
observed_error_count	integer		This speaker’s characteristic ASR failures
last_observed_at	timestamp tz		

6.6 Reporting

ROUTING_DECISION

Why this template was chosen. Auditable, and the training set for the distilled router.

Column	Type	Key / Constraint	Notes
id	uuid	PK	
recording_id	uuid	FK → recording.id	
chosen_template_version_id	uuid	FK → template_version.id	
attached_module_version_ids	uuid []		Composable sub-templates
candidate_set	jsonb	NOT NULL	[{template_version_id, score, reason}] — full shortlist
stage1_filter_source	text	CHECK IN (dicom , lexical_fingerprint , code_word , demographic)	
stage1_candidate_count	smallint		Funnel health metric
confidence	numeric (5,4)	NOT NULL	
orphan_assertion_count	smallint	NOT NULL	Wrong-template signal

required_field_gap_count	smallint	NOT NULL	
was_human_overridden	boolean	NOT NULL DEFAULT false	
human_chosen_template_version_id	uuid	NULL	Gold label for retraining

REPORT_DRAFT

Column	Type	Key / Constraint	Notes
id	uuid	PK	
recording_id	uuid	FK → recording.id	
template_version_id	uuid	FK → template_version.id	
pipeline_run_id	uuid	FK → pipeline_run.id	
rendered_text	text	NOT NULL	Composed narrative
structured_payload	jsonb	NOT NULL	Schema-valid object
overall_confidence	numeric(5,4)	NOT NULL	
flagged_field_count	smallint	NOT NULL	Drives review UI ordering
routing_decision_id	uuid	FK → routing_decision.id	
prompt_bundle_version	text	NOT NULL	Reproducibility
model_versions	jsonb	NOT NULL	Every model id + version used
status	text	CHECK IN (generated , in_review , revised , signed , discarded)	

REPORT_FIELD_VALUE

The atomic unit. Every row must have at least one provenance span or is_grounded = false .

Column	Type	Key / Constraint	Notes
id	uuid	PK	
report_draft_id	uuid	FK → report_draft.id	
template_field_id	uuid	FK → template_field.id	
value_text	text	NULL	
value_numeric	numeric(12,4)	NULL	
value_unit	text	NULL	Normalised (mm, cm, HU)
value_enum	text	NULL	
assertion_status	text	CHECK IN (present , absent , uncertain , not _assessed)	Never a bare boolean
laterality	text	CHECK IN (left , right , bilateral , midline , na) NULL	Isolated for validation
fill_source	text	CHECK IN (dictated , template_default , blanket_normal , inferred , human)	Audit-critical
is_grounded	boolean	NOT NULL	False ⇒ excluded from render, flagged
confidence	numeric(5,4)	NOT NULL	Aggregated — see §8.3.7
is_flagged	boolean	NOT NULL	Surfaced in review UI
flag_reasons	text []		

UNIQUE (report_draft_id, template_field_id). fill_source = 'template_default' on a field the radiologist never mentioned is the single most important thing to make visible in the review UI. It is the system asserting something no human said.

PROVENANCE_SPAN

Invariant I1, made physical.

Column	Type	Key / Constraint	Notes
id	uuid	PK	
report_field_value_id	uuid	FK → report_field_value.id	
transcript_id	uuid	FK → transcript.id	
utterance_id	uuid	FK → transcript_utterance.id	

		NULL	
char_start	integer	NOT NULL	
char_end	integer	NOT NULL	
audio_start_ms	integer	NOT NULL	Powers click-to-listen in review UI
audio_end_ms	integer	NOT NULL	
extraction_confidence	numeric(5,4)		

VERIFICATION_FINDING

Output of validators and the LLM critic.

Column	Type	Key / Constraint	Notes
id	uuid	PK	
report_draft_id	uuid	FK → report_draft.id	
report_field_value_id	uuid	FK NULL	Null for report-level findings
check_id	text	NOT NULL	laterality_consistency , roundtrip_entailment , ...
check_type	text	CHECK IN (rule , schema , llm_critic , roundtrip , cross_modal)	
severity	text	CHECK IN (info , warn , error , block)	block halts autonomy
message	text	NOT NULL	
evidence	jsonb		
resolved_by_iteration	smallint	NULL	Which critic loop pass fixed it
human_verdict	text	CHECK IN (true_positive , false_positive , unreviewed)	Tunes verifier precision

CRITICAL_FINDING_RULE

Detection runs on the **transcript**, not the finished report, so alerts fire before the draft enters any review queue.

Column	Type	Key / Constraint	Notes
id	uuid	PK	
code	text	UNIQUE NOT NULL	PNEUMOTHORAX_TENSION , AORTIC_DISSECTION
finding_label	text	NOT NULL	Human-readable
pattern_type	text	CHECK IN (lexical , regex , llm_semantic)	Lexical first; LLM as a recall net
pattern	text	NOT NULL	
negation_sensitive	boolean	NOT NULL DEFAULT true	“no pneumothorax” must not fire
severity	text	CHECK IN (red , orange)	
sla_minutes	integer	NOT NULL	Communication deadline
escalation_path	jsonb	NOT NULL	Ordered contacts + channels
requires_ack	boolean	NOT NULL DEFAULT true	
approved_by	uuid	FK → app_user.id	Radiologist-owned, not engineer-owned
is_active	boolean	NOT NULL	

CRITICAL_FINDING_ALERT

Column	Type	Key / Constraint	Notes
id	uuid	PK	
recording_id	uuid	FK → recording.id	
rule_id	uuid	FK → critical_finding_rule.id	
utterance_id	uuid	FK → transcript_utterance.id	Provenance
evidence_text	text	NOT NULL	
confidence	numeric(5,4)	NOT NULL	
detected_at	timestamptz	NOT NULL	
sla_due_at	timestamptz	NOT NULL	detected_at + sla_minutes

acknowledged_by	uuid	FK NULL	
acknowledged_at	timestampz	NULL	
is_breach	boolean	NOT NULL DEFAULT false	Computed
outcome	text	CHECK IN (true_positive , false_positive , unreviewed)	Tunes precision

Alerts bypass the normal queue entirely. Tune for **recall at any precision cost** — a false alert wastes thirty seconds; a missed one is the failure this table exists to prevent.

AUTONOMY_CLASS

The unit at which autonomy is granted and revoked.

Column	Type	Key / Constraint	Notes
id	uuid	PK	
code	text	UNIQUE NOT NULL	US_ROUTINE_SIMPLE , XR_CHEST_SCREENING
display_name	text	NOT NULL	
status	text	CHECK IN (not_evaluated , accruing , granted , suspended , revoked)	
baseline_cse_rate	numeric(6,5)	NOT NULL	Measured , not assumed
ni_margin_pp	numeric(4,2)	NOT NULL DEFAULT 1.00	Non-inferiority margin
required_n	integer	NOT NULL	From the power table in §8.3.10
accrued_n	integer	NOT NULL DEFAULT 0	
observed_cse_count	integer	NOT NULL DEFAULT 0	
posterior_prob_ni	numeric(6,5)	NULL	P(non-inferior data)
granted_at / granted_by	timestampz / uuid	NULL	
revoked_at	timestampz	NULL	
revocation_reason	text	NULL	
cusum_statistic	numeric(8,4)	NOT NULL DEFAULT 0	Post-grant monitor
cusum_threshold	numeric(8,4)	NOT NULL	Trip point

AUTONOMY_OBSERVATION

One row per signed report in an accruing or granted class.

Column	Type	Key / Constraint	Notes
id	bigserial	PK	
autonomy_class_id	uuid	FK → autonomy_class.id	
final_report_id	uuid	FK → final_report.id, UNIQUE	
severity_grade	text	CHECK IN (G0 .. G4)	Transcriptionist-assigned, per §5.4.1
is_cse	boolean	NOT NULL	G3 or G4
counted_in_accrual	boolean	NOT NULL	Excludes gold-set overlap
observed_at	timestampz	NOT NULL	

6.7 Review and the flywheel

This is the most commercially valuable data in the system. Design it before you need it.

REPORT_REVISION

Column	Type	Key / Constraint	Notes
id	uuid	PK	
report_draft_id	uuid	FK → report_draft.id	
revision_number	integer	NOT NULL	
revised_by	uuid	FK → app_user.id	
reviewer_role	text	CHECK IN (transcriptionist , radiologist)	Two-layer supervision

structured_payload	jsonb	NOT NULL	State after this revision
rendered_text	text	NOT NULL	
edit_distance_from_prev	integer	NOT NULL	
active_edit_seconds	integer	NOT NULL	Primary value metric input — measure focus time, not wall clock
created_at	timestampz	NOT NULL	

EDIT_EVENT

Field-level diffs. This is your ASR fine-tuning and correction training corpus.

Column	Type	Key / Constraint	Notes
id	bigserial	PK	
report_revision_id	uuid	FK → report_revision.id	
report_field_value_id	uuid	FK NULL	
edit_type	text	CHECK IN (value_change , field_added , field_removed , template_changed , text_rewrite)	
before_value	text		
after_value	text		
audio_start_ms / audio_end_ms	integer	NULL	Links the correction back to audio
transcript_char_start / _end	integer	NULL	
error_category	text	NULL	asr_term , asr_number , laterality , negation , extraction_miss , hallucination , style , template_wrong
severity_grade	text	CHECK IN (G0 , G1 , G2 , G3 , G4) NULL	Assigned by the transcriptionist per the §5.4.1 sampling schedule
is_training_eligible	boolean	NOT NULL DEFAULT true	Excludes eval-set items

Index: (error_category, severity_grade); (is_training_eligible) WHERE is_training_eligible . Every row where error_category = 'asr_term' and the audio span is present is one training example for ASR adaptation. At your volume this reaches useful scale within about 8-12 weeks.

FINAL_REPORT

Column	Type	Key / Constraint	Notes
id	uuid	PK	
study_id	uuid	FK → study.id	
report_draft_id	uuid	FK → report_draft.id	
final_revision_id	uuid	FK → report_revision.id	
signed_by	uuid	FK → app_user.id	
signed_at	timestampz	NOT NULL	
rendered_text	text	NOT NULL	Immutable
structured_payload	jsonb	NOT NULL	
content_hash	text	NOT NULL	Tamper evidence
path_type	text	CHECK IN (transcriptionist_reviewed , radiologist_only)	Autonomy audit
amends_report_id	uuid	FK → final_report.id NULL	Addendum chain
exported_at	timestampz	NULL	
export_status	text	CHECK IN (pending , sent , ack , failed)	HL7/FHIR

6.8 Orchestration and audit

PIPELINE_RUN

--	--	--	--

Column	Type	Key / Constraint	Notes
id	uuid	PK	
recording_id	uuid	FK → recording.id	
pipeline_version	text	NOT NULL	Graph definition version
trigger	text	CHECK IN (upload , manual_retry , backfill , shadow)	
is_shadow	boolean	NOT NULL DEFAULT false	Shadow output never reaches users
status	text	CHECK IN (queued , running , succeeded , failed , cancelled)	
started_at / completed_at	timestampz		
total_cost_usd	numeric(8,4)		Per-report unit economics
error_detail	jsonb	NULL	

STAGE_EXECUTION

Column	Type	Key / Constraint	Notes
id	uuid	PK	
pipeline_run_id	uuid	FK → pipeline_run.id	
stage_name	text	NOT NULL	
attempt	smallint	NOT NULL	
input_ref	jsonb	NOT NULL	Pointers, not payloads
output_ref	jsonb		
model_id / model_version	text	NULL	
prompt_version	text	NULL	
tokens_in / tokens_out	integer		
cost_usd	numeric(8,4)		
duration_ms	integer		
status	text		

AUDIT_LOG

Append-only. Separate database role with INSERT-only grants.

Column	Type	Key / Constraint	Notes
id	bigserial	PK	
actor_id	uuid	NULL	Null for system actions
actor_type	text	CHECK IN (user , system , integration)	
action	text	NOT NULL	
entity_type / entity_id	text / uuid	NOT NULL	
before / after	jsonb		
ip_address	inet		
occurred_at	timestampz	NOT NULL	

6.9 Evaluation

EVAL_SET / EVAL_ITEM / EVAL_RUN / EVAL_RESULT

eval_set — id , name , is_frozen , created_at , description , stratification_spec jsonb

eval_item

Column	Type	Notes
id	uuid PK	
eval_set_id	uuid FK	
recording_id	uuid FK	
gold_transcript_verbatim	text	Disfluencies retained
gold_utterance_labels	jsonb	
gold_template_version_id	uuid FK	

gold_structured_payload	jsonb	
gold_codeword_occurrences	jsonb	
audio_quality_bucket	text	clean / moderate / noisy — stratification
annotated_by	uuid FK	

eval_run — id, eval_set_id, pipeline_version, config_snapshot jsonb, started_at, metrics jsonb (all \$5.2 metrics), is_release_gate boolean

eval_result — id, eval_run_id, eval_item_id, metric_key, metric_value numeric, error_detail jsonb

Release gate: no pipeline version reaches production unless an eval_run with is_release_gate = true shows no regression on CSE_DRAFT, HALLUC_RATE, or ROUTE_TOP1 per template.

6.10 Onboarding and corpus import

IMPORT_BATCH

One execution of the onboarding pipeline, or of any later re-run.

Column	Type	Key / Constraint	Notes
id	uuid	PK	
batch_type	text	CHECK IN (template, report_corpus, paired_audio, shorthand, roster, boilerplate, critical_rules)	
trigger	text	CHECK IN (initial_onboarding, template_change, new_radiologist, periodic_remine, site_expansion)	Not one-time
stage	text	NOT NULL	Which onboarding stage (\$7.8)
status	text	CHECK IN (uploading, parsing, awaiting_review, approved, rejected, applied)	
submitted_by / approved_by	uuid	FK → app_user.id	Approval is a clinical act
item_count / accepted_count / rejected_count	integer		
blocking_issue_count	integer	NOT NULL	Non-zero blocks applied
applied_at	timestamptz	NULL	

IMPORT_ARTIFACT

Each uploaded file, with parse provenance.

Column	Type	Key / Constraint	Notes
id	uuid	PK	
import_batch_id	uuid	FK → import_batch.id	
original_filename	text	NOT NULL	
content_hash	text	NOT NULL	Idempotency — re-uploading the same file is a no-op
mime_type	text		.docx, .rtf, .txt, .csv, .wav, RIS export
parse_status	text	CHECK IN (ok, partial, failed)	
parse_warnings	jsonb		

UNIQUE (import_batch_id, content_hash) .

TEMPLATE_IMPORT_CANDIDATE

Parsed but **not yet live**. Templates never go straight into template_version .

Column	Type	Key / Constraint	Notes
id	uuid	PK	
import_batch_id	uuid	FK	
template_id	uuid	FK	

import_artifact_id	uuid	FK	
proposed_code	text		
proposed_json_schema	jsonb	NOT NULL	LLM-parsed from the source document
proposed_sections	jsonb		
proposed_modality / proposed_body_region	text		
proposed_spoken_study_code	text		
parse_confidence	numeric(5,4)		Low confidence ⇒ mandatory field-by-field review
review_status	text	CHECK IN (pending , approved , edited , rejected , merged)	
merged_into_template_id	uuid	FK NULL	Dedup outcome
promoted_template_version_id	uuid	FK NULL	Set on approval

TEMPLATE_MERGE_PROPOSAL

Near-duplicate detection. Merging before routing is trained is far cheaper than after.

Column	Type	Notes
id	uuid PK	
import_batch_id	uuid FK	
template_a_id / template_b_id	uuid FK	
similarity_score	numeric(5,4)	Field-set Jaccard + text similarity
differing_fields	jsonb	Shown to the reviewer
decision	text CHECK IN (merge , keep_separate , pending)	
decided_by	uuid FK	Radiologist

CORPUS_REPORT

Historical signed reports with no matching audio. Your largest asset.

Column	Type	Key / Constraint	Notes
id	uuid	PK	
import_batch_id	uuid	FK	
external_report_id	text	UNIQUE	Source-system key
report_text	text	NOT NULL	
report_date	date		
radiologist_id	uuid	FK NULL	Enables per-speaker priors
referring_doctor	text	NULL	Referrer→template prior
patient_sex / patient_age_years	text / smallint		Demographic priors
embedding	vector(768)		Exemplar retrieval for routing and few-shot
is_deidentified	boolean	NOT NULL	Gate before any external API call

CORPUS_REPORT_TEMPLATE_MAP

Derived, not supplied. This mapping usually does not exist in the source data.

Column	Type	Notes
id	uuid PK	
corpus_report_id	uuid FK	
template_id	uuid FK	
match_method	text CHECK IN (explicit , structural , llm , human)	Explicit if the source recorded it
confidence	numeric(5,4)	
is_verified	boolean	Human-confirmed sample

Feeds three things at once: the **usage histogram** (power-law head detection), the **routing exemplar index**, and the **referrer prior**. Sample-verify at least 200 mappings by hand before trusting the histogram.

LEXICON_MINING_RUN / COLLISION_AUDIT_FINDING

lexicon_mining_run — id, import_batch_id, corpus_size, terms_extracted, terms_new, terms_auto_accepted, terms_pending_review, run_at

collision_audit_finding — the gate that catches LMC vs LMP

Column	Type	Notes
id	uuid PK	
term_a_id / term_b_id	uuid FK → lexicon_term.id	
phonetic_distance	numeric (5,4)	
collision_class	text CHECK IN (e_set_letter , homophone , near_homophone , natural_word_overlap)	
severity	text CHECK IN (block , warn)	block if both map to different templates or opposite clinical meanings
resolution	text CHECK IN (renamed , accepted_with_margin_guard , pending)	
resolved_by	uuid FK	

Runs over both lexicon_term shorthand and template_version.spoken_study_code . A block -severity finding prevents import_batch.status = 'applied' .

BOILERPLATE_CANDIDATE

Mined normal statements, proposed for template_field.default_normal_text .

Column	Type	Notes
id	uuid PK	
template_field_id	uuid FK	
candidate_text	text	
corpus_frequency	integer	How often this exact sentence appears for this field
corpus_share	numeric (5,4)	As a fraction of reports using the template
review_status	text CHECK IN (pending , approved , rejected)	

This table is absence_policy Pass 2 (§8.3.6): rank by corpus_share , present the top ~40, let the radiologist promote.

ONBOARDING_READINESS_CHECK

The gate between onboarding and pilot.

Column	Type	Notes
id	uuid PK	
check_id	text	corpus_template_coverage , collision_audit_clear , voice_enrollment_complete , ...
status	text CHECK IN (pass , warn , fail)	
measured_value / threshold	numeric	
detail	jsonb	
evaluated_at	timestampz	

6.11 Cross-cutting data policies

Concern	Policy
PHI at rest	Column-level encryption on patient.mrn , patient.name_enc . Postgres TDE or pgcrypto
PHI in prompts	Patient name never sent to any external API. Use patient.pseudonym . Age and sex are sent (clinically necessary for routing/validation)
Voice biometrics	radiologist_profile.voice_embedding is sensitive personal data under India's DPDP Act 2023. Explicit consent, recorded in voice_consent_ref
Audio retention	Indefinite, lossless FLAC. ~7.7 MB per 4-min WAV → ~560 GB/yr at 200/day, ~280 GB/yr as FLAC. Storage cost is negligible against fine-tuning value

Audio codec	FLAC or WAV only. Lossy compression is rejected at ingest — it destroys the spectral detail ASR fine-tuning depends on, irreversibly
Dual legal basis	Clinical-record retention is statutory. Model-training retention is a separate purpose under DPDP purpose-limitation and needs its own basis and disclosure. Keep two stores: the clinical archive, and a pseudonymised training corpus gated by <code>recording.is_training_corpus_eligible</code>
Right to erasure	Cascade defined: <code>patient</code> → pseudonymise; audio purged; <code>final_report</code> retained per medical records law (erasure request cannot override statutory retention)
Backups	PITR, encrypted, tested restore quarterly
Partitioning	<code>asr_segment</code> , <code>edit_event</code> , <code>audit_log</code> partitioned monthly

6.12 Model adaptation

VERBATIM_TRANSCRIPT

Token-aligned ground truth. The only asset that can train ASR.

Column	Type	Key / Constraint	Notes
<code>id</code>	<code>uuid</code>	PK	
<code>recording_id</code>	<code>uuid</code>	FK → <code>recording.id</code>	
<code>text</code>	<code>text</code>	NOT NULL	Disfluencies, false starts, and corrections retained
<code>source</code>	<code>text</code>	CHECK IN (<code>human_annotation</code> , <code>verified_correction</code> , <code>imported</code>)	<code>verified_correction</code> = reviewer-confirmed ASR output
<code>annotator_id</code>	<code>uuid</code>	FK → <code>app_user.id</code>	
<code>includes_disfluencies</code>	<code>boolean</code>	NOT NULL	False ⇒ not training-eligible
<code>audio_duration_seconds</code>	<code>numeric(8,2)</code>	NOT NULL	Corpus-hours accounting
<code>capture_device_class</code>	<code>text</code>	NOT NULL	Denormalised — the filter that matters most
<code>is_eval_set_member</code>	<code>boolean</code>	NOT NULL	True ⇒ permanently excluded from training
<code>quality_checked</code>	<code>boolean</code>	NOT NULL DEFAULT false	

TRAINING_CORPUS_SNAPSHOT

An immutable, reproducible selection. Training runs reference a snapshot, never a live query.

Column	Type	Key / Constraint	Notes
<code>id</code>	<code>uuid</code>	PK	
<code>name</code>	<code>text</code>	NOT NULL	
<code>purpose</code>	<code>text</code>	CHECK IN (<code>asr_global</code> , <code>asr_speaker</code> , <code>utterance_classifier</code> , <code>router</code>)	
<code>filter_spec</code>	<code>jsonb</code>	NOT NULL	Exact selection criteria
<code>item_count</code>	<code>integer</code>	NOT NULL	
<code>total_audio_hours</code>	<code>numeric(8,2)</code>	NOT NULL	Checked against \$8.6.5 gates
<code>speaker_distribution</code>	<code>jsonb</code>	NOT NULL	Guards against single-speaker dominance
<code>device_class_filter</code>	<code>text[]</code>	NOT NULL	
<code>excludes_eval_set</code>	<code>boolean</code>	NOT NULL CHECK (<code>excludes_eval_set</code> = true)	Enforced at the schema level
<code>content_hash</code>	<code>text</code>	UNIQUE NOT NULL	Reproducibility

The `CHECK` constraint on `excludes_eval_set` is deliberate. Eval-set leakage is silent, unrecoverable, and invalidates every number downstream of it. Make it impossible in the schema rather than a rule someone has to remember.

MODEL_ADAPTATION_RUN

Column	Type	Key / Constraint	Notes
<code>id</code>	<code>uuid</code>	PK	
<code>snapshot_id</code>	<code>uuid</code>	FK → <code>training_corpus_snapshot.id</code>	

target	text	CHECK IN (asr_global , asr_speaker , utterance_classifier , router)	
method	text	CHECK IN (azure_custom_speech , whisper_lora , full_ft , distillation)	
base_model / base_version	text	NOT NULL	
radiologist_id	uuid	FK NULL	Per-speaker adapters
hyperparams	jsonb		
baseline_metrics	jsonb	NOT NULL	Gold-set scores before adaptation
result_metrics	jsonb	NULL	Same metrics after
relative_improvement	numeric(6,4)	NULL	Negative ⇒ auto-reject
prerequisite_gates_passed	jsonb	NOT NULL	\$8.6.5 checklist result
status	text	CHECK IN (training , evaluated , shadow , promoted , rejected)	
artifact_ref	text	NULL	
promoted_at	timestampz	NULL	

6.13 Tenant isolation (built from day one, per D19)

Decision, revised. Real tenant isolation is enforced from the start, not deferred to a second client signing. The heavier structural step — a dedicated per-tenant data space (separate schema or partition set) — is planned as a later, client-managed migration once the tenant count justifies it; that migration and its zero-downtime execution are the client’s own responsibility to plan and run.

Built now — enforcement, not just a column:

Table	Isolation column	Enforcement
patient , study , recording	tenant_id uuid NOT NULL	Row-level security policy + mandatory application-layer scoping
template , template_version	tenant_id uuid NOT NULL	Same — no cross-tenant template sharing by default
lexicon_term , lexicon_set	tenant_id uuid NULL	Nullable only for a global starting lexicon; tenant-specific terms are isolated
app_user , radiologist_profile	tenant_id uuid NOT NULL	A user session is scoped to exactly one tenant at login
final_report , report_draft	tenant_id uuid NOT NULL (denormalised from study)	RLS-enforced; this is the table where a leak would matter most
import_batch and all \$6.10 onboarding tables	tenant_id uuid NOT NULL	Onboarding runs per tenant, isolated from the start

A tenant table (id , name , status , created_at) anchors the column. Two enforcement layers, both live from day one:

- Application layer.** Every query path runs through a mandatory tenant-scoping wrapper — there is no code path that queries these tables without a tenant_id filter. Enforced by convention plus a lint/test rule, not by developer discipline alone.
- Database layer.** Postgres row-level security (RLS) policies on every table above, keyed to the session’s tenant_id . This is the layer that holds even if an application-layer bug slips through — the database refuses to return rows outside the current tenant regardless of what the query asked for.

At one tenant this has no visible effect and negligible overhead — every row belongs to the same tenant, every query already scopes to it. The isolation is real starting now; it simply has nothing to isolate against yet.

What is still deferred, and correctly so:

- Per-tenant billing/metering
- Per-tenant admin console or self-service onboarding
- Cross-tenant analytics or benchmarking features
- Physical data-space separation (per-tenant schema or partition) — planned as the structural evolution once tenant count justifies it, executed by the client as a live migration when that time comes
- Any UI concept of “switching tenants”

These remain gated on a second client actually signing, not built speculatively now.

6.14 Model configuration (admin panel)

Backs §7.9.6. Deliberately small — this is a configuration surface, not a model registry product.

MODEL_PROVIDER

Canonical list of providers the panel can point a task at. Seeded with the entries from §7.9.2 and §7.9.4; new rows added as new providers are tried.

Column	Type	Key / Constraint	Notes
id	uuid	PK	
name	text	UNIQUE NOT NULL	anthropic , openai , google , moonshot , local_openai_compatible
kind	text	CHECK IN (cloud_api , local_openai_compatible)	Determines which fields below apply
default_endpoint	text	NULL	Provider's standard API base URL; local providers set this per-deployment instead
auth_method	text	CHECK IN (api_key , none)	Local endpoints on trusted hardware may need none
is_active	boolean	NOT NULL DEFAULT true	

MODEL_DEFINITION

A specific model offered by a provider, with its pricing — the thing a task gets assigned to.

Column	Type	Key / Constraint	Notes
id	uuid	PK	
provider_id	uuid	FK → model_provider.id	
model_identifier	text	NOT NULL	Exact pinned string, e.g. claude-sonnet-5-20260415 — never an alias like latest (\$8.5)
display_name	text	NOT NULL	Claude Sonnet 5 , Llama 3.1 8B (local)
input_price_per_1k	numeric(10,6)	NULL	Cloud: real vendor rate. Local: 0
output_price_per_1k	numeric(10,6)	NULL	Same
endpoint_override	text	NULL	For local models — this deployment's specific workstation/server address
context_window	integer	NULL	Informational
is_active	boolean	NOT NULL DEFAULT true	

UNIQUE (provider_id, model_identifier) . Local models are just rows here with provider_id pointing at the local_openai_compatible provider and their own endpoint_override — no separate code path from cloud models.

TASK_MODEL_ASSIGNMENT

The current live assignment, one row per task, plus its full history.

Column	Type	Key / Constraint	Notes
id	uuid	PK	
task_key	text	NOT NULL	extraction , self_correction , verification , routing_pick , utterance_classification , routing_shortlist , roundtrip_check — the \$7.9.3 task list
task_bucket	text	CHECK IN (consequential , bounded)	Determines how much testing is required before activation
model_definition_id	uuid	FK → model_definition.id	
status	text	CHECK IN (proposed , testing , active , retired)	
eval_run_id	uuid	FK → eval_run.id NULL	The gold-set test result that approved this assignment — required before status can become active
estimated_cost_per_report	numeric(10,4)	NULL	Computed from the model's pricing and the task's typical token volume, shown in the panel
activated_at	timestampz	NULL	
activated_by	uuid	FK → app_user.id NULL	
retired_at	timestampz	NULL	

Partial unique index on `(task_key) WHERE status = 'active'` — exactly one live assignment per task at any time. A proposed assignment cannot reach `active` without a non-null `eval_run_id`, enforced at the application layer (Postgres `CHECK` cannot reference another table, so this is one of the few rules that lives in code rather than schema — flag it in code review accordingly).

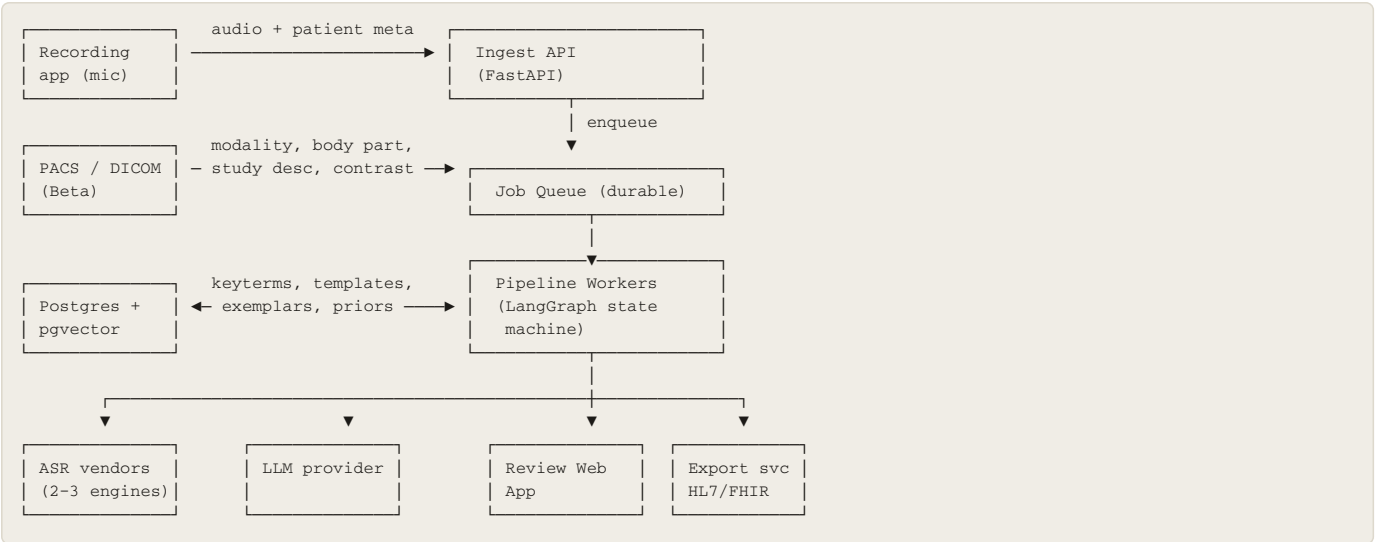
TASK_MODEL_ASSIGNMENT_LOG

Append-only audit trail, since `task_model_assignment` only holds current + one prior state per row lifecycle.

Column	Type	Notes
id	bigserial PK	
task_key	text	
model_definition_id	uuid FK	
event	text CHECK IN (proposed, tested, activated, retired)	
eval_run_id	uuid FK NULL	
actor_id	uuid FK	
occurred_at	timestampz NOT NULL	

7. High-Level Design

7.1 System context

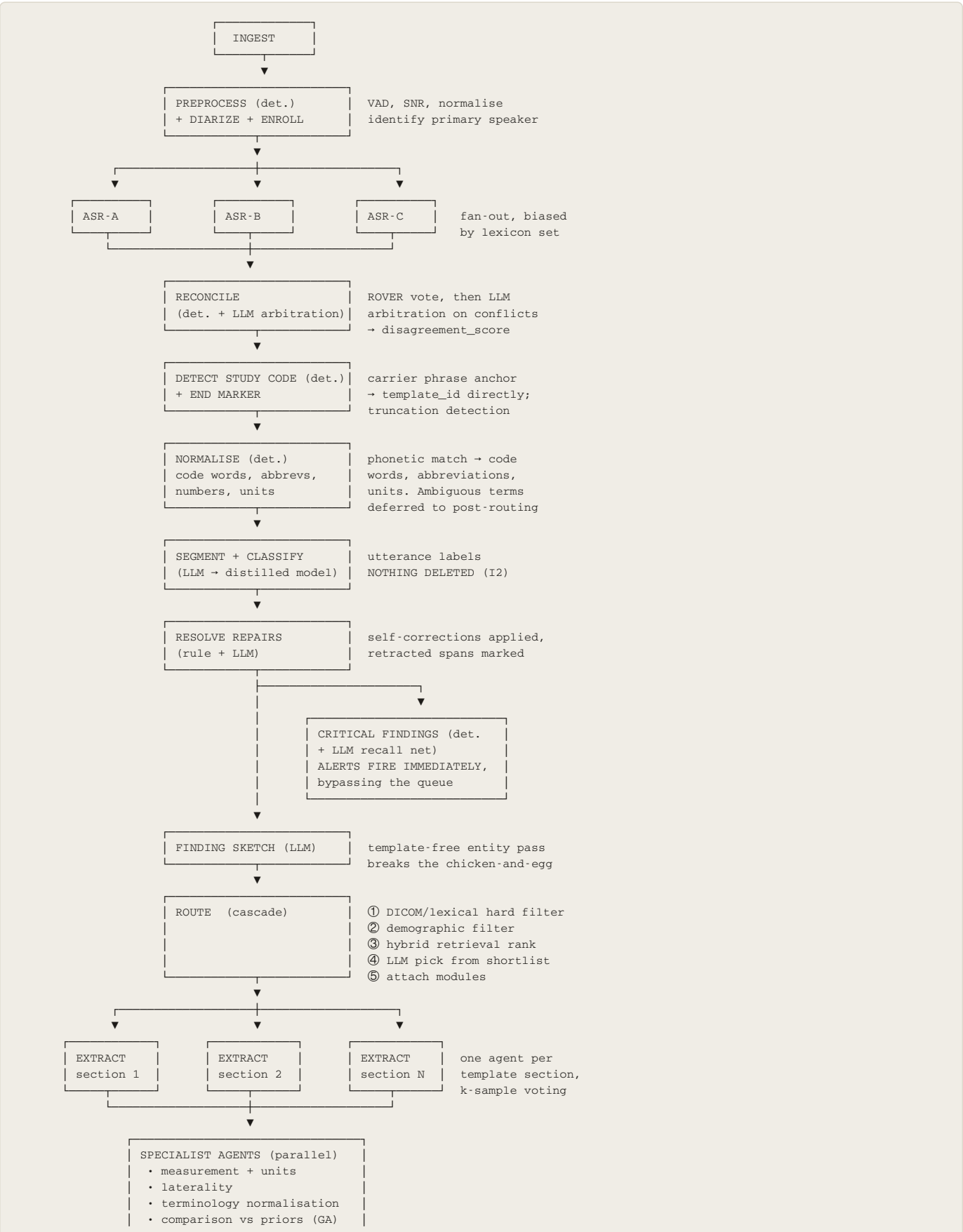


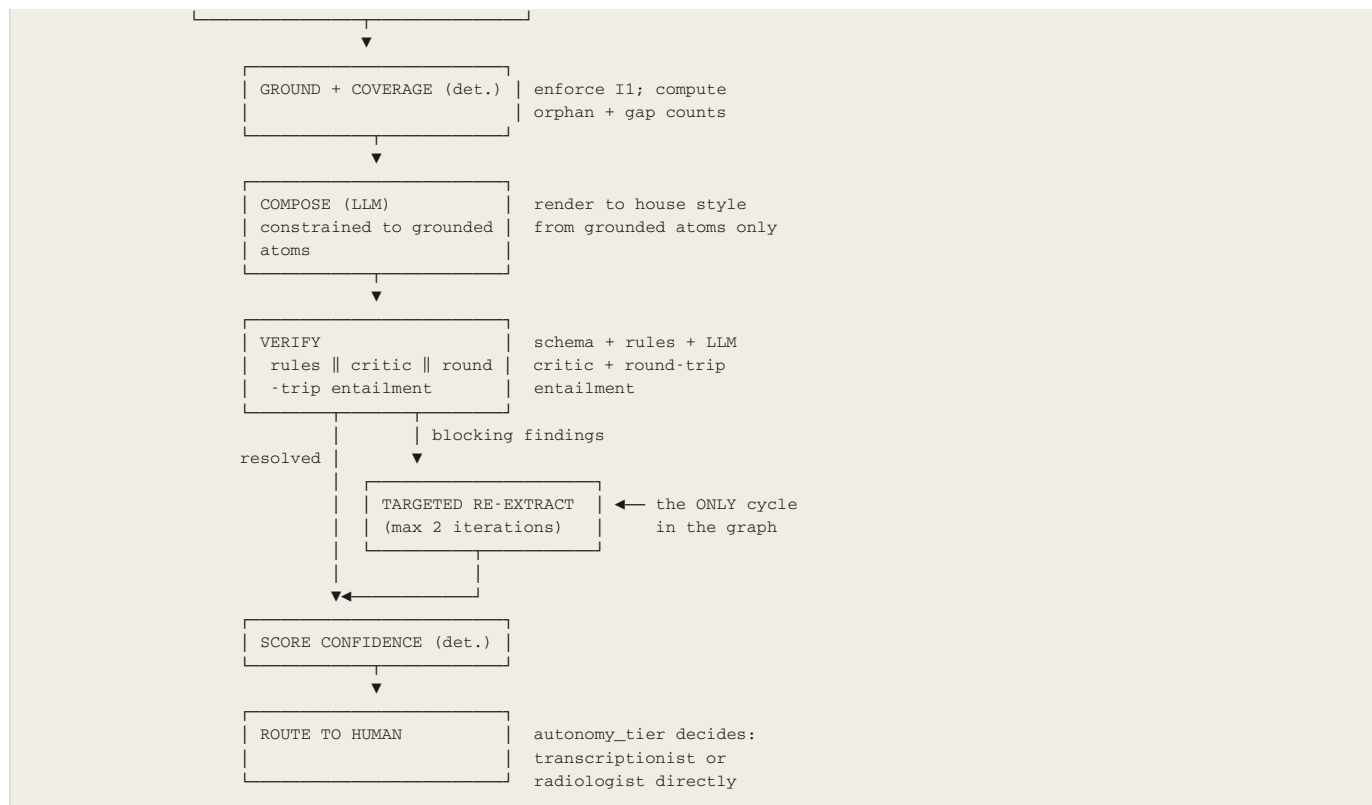
7.2 Component responsibilities

Component	Responsibility	Notes
Ingest API	Accept upload, validate audio, compute <code>content_hash</code> + SNR, create <code>study / recording</code> , enqueue	Rejects duplicates via hash; warns on low sample rate
DICOM Sync (Beta)	Poll/subscribe for study metadata; join on accession or (MRN, <code>study_date</code>); enrich <code>study</code>	Nightly batch is sufficient; C-FIND or DICOMweb QIDO-RS
Job Queue	Durable, priority-aware (<code>stat</code> > <code>urgent</code> > <code>routine</code>), retry with backoff	Survives vendor outage
Pipeline Workers	Execute the agent graph; persist every stage	Horizontally scalable; idempotent per (<code>recording_id</code> , <code>pipeline_version</code>)
Knowledge Service	Serve lexicons, templates, exemplars, priors; hybrid BM25 + vector retrieval	Read-heavy; cacheable
Review Web App	Transcriptionist and radiologist editing, click-to-listen, flag triage, edit capture	The product surface that determines adoption
Export Service	HL7 v2 ORU^R01 / FHIR <code>DiagnosticReport</code> ; retry + ack tracking	GA
Knowledge Onboarding Console	Import and version templates, report corpus, shorthand, paired audio, rosters, boilerplate, critical-finding rules. Staged, gated, re-runnable	\$7.8 — the scaling mechanism if this ever runs at a second lab
Model Config Admin Panel	Assign a model (cloud or local) per pipeline task; cost	\$7.9.6 — the cost-optimization tool for \$15.0a

	estimate per assignment; mandatory gold-set test before any swap goes live	
Eval Service	Run gold set against any pipeline version; release gate	Blocks deploys on regression
Observability	Traces (Langfuse), metrics, cost per report, drift alerts	

7.3 The agent graph





7.4 What is agentic and what is not

Node	Type	Rationale
Preprocess, diarize	Deterministic	Signal processing
ASR fan-out	Deterministic orchestration, model inference	
Reconcile	Hybrid — ROVER vote, LLM only on conflicts	LLM sees only disputed spans, not the whole transcript
Normalise	Deterministic (phonetic index)	Reproducible, fast, auditable
Segment + classify	LLM at V1 → distilled encoder at GA	Distillation once ~5k labelled utterances exist
Resolve repairs	Rule-triggered, LLM-resolved	High clinical stakes; both layers required
Finding sketch	LLM	Cheap, template-free
Route	Cascade: rules → retrieval → LLM	Rules first; LLM only picks from a shortlist
Extract	LLM, parallel per section, k-sample self-consistency	Latency budget spent here
Specialists	LLM + deterministic validators	Measurement parsing is rule-checked
Ground + coverage	Deterministic	Invariant enforcement must not be probabilistic
Compose	LLM, input restricted to grounded atoms	Cannot introduce ungrounded content
Verify	Rules LLM critic entailment	Precedent: GPT-4 achieved 100% recall / 76.9% precision on clinically significant SR errors — tune for recall
Confidence	Deterministic formula	

Explicitly rejected: an autonomous ReAct loop that plans its own steps. Non-determinism is a liability in a clinical pipeline. The same input must produce the same output, and every hop must be reconstructible from `stage_execution`.

7.5 Technology decisions

Layer	Choice	Rationale	Alternative considered
Language	Python 3.12	Ecosystem for ASR/LLM/eval	—
API	FastAPI + Pydantic v2	Schema types shared with LLM structured output	—
Orchestration	LangGraph inside workers	Explicit state machine, checkpointing, bounded cycles	Plain DAG (viable); Temporal (better durability, higher ops cost)
Queue	Redis + Celery (V1) → Temporal (GA)	V1 simplicity; GA needs durable replay for archive reprocessing	SQS/RabbitMQ
DB	Postgres 16 + pgvector	One system for relational + vector; on-prem friendly	Qdrant if vector volume outgrows pgvector
Retrieval	Hybrid BM25 (<code>tsvector</code>) + dense	Exact term match matters as much as	Pure dense — rejected

		semantics here	
Structured output	Provider JSON-schema mode; Outlines/XGrammar for open models	Guarantees schema validity	Free-form + parse — rejected
ASR	Benchmark: Deepgram Nova-3 Medical, Azure Speech custom, Whisper (medium and large-v3 both)	See §7.6 warning	AssemblyAI, ElevenLabs Scribe, Parakeet
Diarization	Vendor built-in, or pyannote + speaker enrollment	Enrollment beats clustering	
Phonetics	Double Metaphone + weighted Levenshtein on phonetic key	Handles E-set confusions	Soundex — too coarse
Object storage	S3-compatible, SSE-KMS		
Tracing	Langfuse	Prompt versioning + cost attribution	LangSmith
Serving	Docker + K8s or ECS		

7.6 ASR selection warning

Do not assume the largest model is best. Published results on Indian-accented speech show Whisper-large-v3 reaching 19.0% WER against Whisper-medium’s 13.2%, with large-v3’s insertion rate spiking to 9.62% so that hallucinated insertions become 50.7% of all errors. Meanwhile the Svarah benchmark put Whisper-large at 7.2% WER on Indian-accented speech versus roughly 21% for Azure and 20-30% for Google, though even Whisper-large only reached 11.2% on utterances dense with named entities.

Two conclusions: 1. **Benchmark medium against large-v3 on your own audio.** Track `INS_RATE` separately from `WER` — inserted text is the failure mode that fabricates findings. 2. **Vendor results vary enormously by dataset.** Your gold set is the only authority.

For domain vocabulary specifically, Deepgram reports a 6.79% keyword error rate on medical terminology for Nova-3 Medical, which is the right order of magnitude to expect before any biasing or fine-tuning.

7.7 Deployment topology

Environment	Purpose	Data
dev	Development	Synthetic + de-identified
staging	Eval runs, release gates	Gold set only
shadow	New pipeline versions run in parallel on live traffic; output persisted, never surfaced	Live
prod	Live	Live

Every release: staging eval gate → shadow for ≥500 reports → canary on one template class → full rollout. Rollback is a config flag on `pipeline_version`, not a redeploy.

7.8 Knowledge Onboarding Console

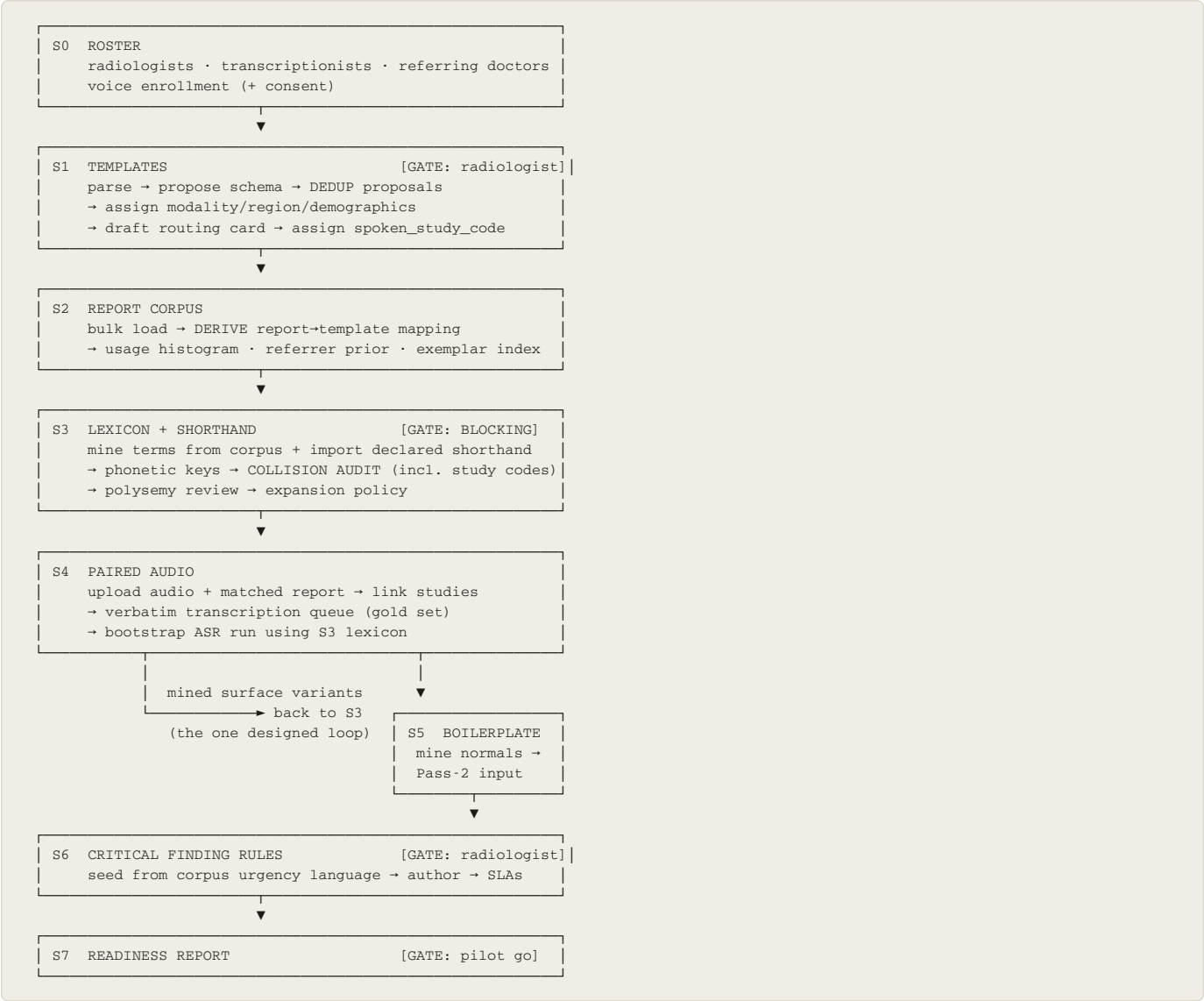
The module that loads templates, the report corpus, shorthand, paired audio, rosters, boilerplate, and critical-finding rules into the system.

Built as a versioned, re-runnable pipeline — not a one-time import script. Re-run triggers arrive within weeks of go-live:

Trigger	What re-runs
Template revised or added	Parse → dedup check → collision audit → exemplar re-index
New radiologist joins	Voice enrollment; speaker profile bootstrapped from her historical reports; personal shorthand
New shorthand appears organically	Detected from <code>edit_event</code> , proposed for lexicon review
Corpus grows	Periodic re-mining picks up terms that didn’t exist at onboarding
ASR improved	Archive reprocessing — the reason audio is retained indefinitely
Second lab deployment	The whole pipeline. This module is the scaling mechanism

7.8.1 STAGE DEPENDENCIES

The imports depend on each other, and one dependency is circular. A screen with five file pickers will not work.



The S3→S4 loop is deliberate. Surface variants for shorthand ("you gee ess" , "U G S" , "yugs") can only be mined from real transcripts, which need ASR, which needs the lexicon for biasing. Bootstrap: run S3 with declared shorthand only → run S4 ASR → mine observed variants → re-run S3 to populate `lexicon_surface_variant` . Two passes, then it converges.

7.8.2 STAGE DETAIL

Stage	Inputs accepted	Automated	Human gate
S0 Roster	CSV, or HR/RIS export	Deduplicate; match to existing users	Consent capture for voice biometrics
S1 Templates	.docx , .rtf , .txt , RIS export	LLM parses to JSON Schema; clusters near-duplicates; drafts routing cards; proposes study-code phrases	Radiologist approves each schema, each merge, each study-code phrase. Low <code>parse_confidence</code> forces field-by-field review
S2 Corpus	Bulk text/CSV export of signed reports	Derives report→template mapping; builds histogram, referrer prior, exemplar index	Verify ≥200 mappings by hand before trusting the histogram
S3 Lexicon	Declared shorthand list + mined terms	Phonetic keys; collision audit ; polysemy flagging	Blocking: every <code>severity</code> = 'block' collision resolved before apply
S4 Paired audio	Audio files + matched reports	Links studies; queues verbatim transcription; bootstrap ASR; mines surface variants	Transcriptionists produce verbatim transcripts

S4 does not train anything. Uploaded pairs become the frozen gold set, mined surface variants, retrieval exemplars, and threshold calibration data. Signed reports are not token-aligned to audio, and the legacy recordings are from the wrong microphones. See §8.6 for why, and for when adaptation actually becomes possible. | **S5 Boilerplate** | Derived from S2 | Ranks candidate normal statements by `corpus_share` | Radiologist promotes to `default_normal` (Pass 2) | | **S6 Critical rules** | Radiologist-authored, corpus-seeded | Suggests candidates from urgency language in corpus | **Radiologist authors and approves.** SLAs and escalation contacts | | **S7 Readiness** | — | Runs all checks | Pilot go/no-go |

7.8.2B BASELINE ERROR-RATE AUDIT (D12)

Folded into the onboarding flow as a formal step, not a standalone task — run once per client, before the pilot begins.

- 1. Pull 50 reports the client has already signed and delivered (unrelated to the pilot — this measures the client’s *current* process).
- 2. A transcriptionist compares each final report against its original dictation audio and grades it `no error` / `minor error` / `significant error` using the same `G0 - G4` rubric (\$5.4).
- 3. `baseline_cse_rate` = (count of significant errors) ÷ 50, stored on the client’s `autonomy_class` rows (\$6.12) and used throughout §8.3.10’s non-inferiority calculations.

This runs during **S2 (Report Corpus)** of the console (§7.8.1), since it needs the same historical-report access as corpus loading, and its output gates **S7 readiness** — no pilot go-live without a measured baseline.

7.8.3 READINESS CHECKS

check_id	Threshold	Severity
corpus_template_coverage	≥90% of historical reports map to a loaded template	fail
collision_audit_clear	Zero unresolved <code>block</code> findings	fail
study_code_uniqueness	Every active template has a unique, phonetically separated spoken code	fail
critical_field_signoff	Every pilot template has <code>is_critical</code> marked (Pass 1)	fail
voice_enrollment_complete	All pilot radiologists enrolled with consent recorded	fail
gold_set_frozen	Both partitions annotated and frozen	fail
critical_rules_approved	≥1 approved rule with escalation path and SLA	fail
template_dedup_reviewed	All merge proposals decided	warn
lexicon_size	≥2,000 mined terms	warn
exemplar_coverage	≥20 exemplars per pilot template	warn

7.8.4 DESIGN RULES

- **Nothing goes live without approval.** Parsed templates land in `template_import_candidate`, never directly in `template_version`. LLM parsing of a clinical template is a proposal, not a decision.
- **Idempotent by content hash.** Re-uploading the same file is a no-op, so a partially failed import can be retried safely.
- **Every import is versioned and reversible.** `import_batch` records what changed; applying a batch creates new `template_version` rows rather than mutating existing ones.
- **De-identify before any external call.** `corpus_report.is_deidentified` gates LLM-assisted parsing and mapping.
- **The console is the interface for weeks 1-5 of §12.** That work is not scripts run beside the product; it is the product’s first module, exercised by the people who will operate it.

7.9 Model Selection: Frontier vs. Open-Source

This section consolidates every model decision discussed for the pipeline: which frontier LLM, which ASR engine, and which open-source model runs locally — with pricing, the reasoning behind the split, and the guardrails around changing any of it.

7.9.1 THE GOVERNING RULE

Split by consequence, not by cost. Every task in the agent graph (§7.3) falls into one of two buckets:

- **Consequential tasks** — a mistake becomes a wrong clinical report. Extraction, self-correction resolution, verification, final routing pick. These use the strongest model affordable, regardless of price, because the cost of an error here is not measured in rupees.
- **Bounded tasks** — the model chooses from a short, well-defined set of options. Utterance classification, routing shortlist ranking, round-trip comparison. A small model does these about as well as a large one, so the cheapest reliable option wins.

This rule, not the price table below, is what should survive any future re-pricing. Prices move constantly; the consequence of each task does not.

7.9.2 FRONTIER MODEL COMPARISON (CLOUD, PAID)

Pricing per million tokens, converted at ₹96/USD. **Verify current rates before committing to a contract — this market moves within weeks.**

Provider	Model	Input \$/M	Output \$/M	₹ per report*
Anthropic	Claude Opus 5	\$5.00	\$25.00	~39
Anthropic	Claude Sonnet 5	\$3.00	\$15.00	~23
Anthropic	Claude Haiku 4.5	\$1.00	\$5.00	~8
OpenAI	GPT-5.4	\$2.50	\$15.00	~21
OpenAI	GPT-5.6 Terra	\$2.00	\$12.00	~17
OpenAI	GPT-5.6 Luna	\$0.20	\$1.20	~2

Google	Gemini 3.1 Pro	\$2.00	\$12.00	~17
Google	Gemini 3.5 Flash	\$1.50	\$9.00	~12
Google	Gemini 3.1 Flash-Lite	\$0.25	\$1.50	~2
Moonshot AI	Kimi K2 Thinking	\$0.60	\$2.50	~4

*Assuming ~54,000 input words and ~5,400 output words per report, across the full set of LLM calls the pipeline makes for one dictation (extraction across sections, self-consistency samples, classification, routing, verification).

Recommendation: Claude Sonnet 5 for consequential tasks, Claude Haiku 4.5 for bounded tasks. Reasoning:

- On the published price table alone, Sonnet is mid-pack — GPT-5.4 and Opus cost more, several others cost less. It is not the expensive choice.
- Claude has tested strongest in general use at following detailed instructions precisely and staying inside a strict output format, which is the core requirement of template-constrained extraction (§8.3.5).
- Keeping both tiers on one provider simplifies billing, version pinning (§8.5), and the release-gate process — one vendor’s deprecation calendar to track instead of several.
- Haiku costs roughly a third of Sonnet and is adequate for every bounded task, so it captures most of the available saving without touching the tasks where accuracy matters.

Kimi K2 Thinking is the one outside candidate worth testing. At ~₹4/report it is far cheaper than Sonnet and has shown strong general reasoning benchmarks. It has not been tested on Indian-accented radiology dictation by anyone, including this design. Add it to the week-3 bake-off (§12) as a second candidate for consequential tasks, evaluated on the gold set exactly like the ASR engines are — but do not adopt it on price alone.

Do not adopt a new model on name recognition. Provider naming is not a reliable signal of what a model is. Meta’s “Muse Spark,” despite following Meta’s history of open releases (Llama), is **proprietary and API-only — not open source, not locally hostable**. Before adding any model to this comparison, confirm licensing and hosting terms directly; do not infer them from the model family’s past reputation.

7.9.3 RECOMMENDED PER-TASK MODEL ASSIGNMENT

Task	Bucket	Model	Rationale
Finding extraction (audio → structured fields)	Consequential	Claude Sonnet 5	Highest accuracy need
Self-correction resolution	Consequential	Claude Sonnet 5	High stakes, needs strong language understanding
Final verification / critic pass	Consequential	Claude Sonnet 5	The safety net — do not weaken it
Template routing, final pick from shortlist	Consequential	Claude Sonnet 5	Wrong pick corrupts the whole report structure
ASR (speech-to-text)	—	Vendor per \$7.6 bake-off	Specialized ASR beats general LLMs at this job
Utterance classification	Bounded	Claude Haiku 4.5, or local (see §7.9.4)	Simple labeling
Routing shortlist ranking	Bounded	Claude Haiku 4.5, or local	Ranking a short list, not open-ended judgment
Round-trip verification check	Bounded	Claude Haiku 4.5, or local	Comparison task
Code word / shorthand matching	—	Rule-based phonetic index (§8.3.2)	No LLM needed — already solved, cheaper and more reliable

Blended LLM cost at this split: roughly **₹15-17 per report**, against ₹23 for an all-Sonnet pipeline — a 30-40% cut with no change to the tasks that carry clinical risk.

7.9.4 LOCAL OPEN-SOURCE MODELS (RADIOLOGIST WORKSTATION)

For the bounded tasks, running locally on existing radiologist hardware (§8.6.4 discussion) removes their cost entirely rather than just discounting it, provided the workstation has spare capacity and IT approves a background service on a PACS-connected machine.

Model	Size	License	Fit
Llama 3.1 / 3.3	8B	Open weights	Recommended — classification, routing, comparison
Qwen 2.5	7B	Open weights	Recommended alternative — strong at short-list ranking
Mistral / Mixtral	7B-8x22B	Open weights	Viable, less tested for this task set than the two above
DeepSeek-V3	671B	Open weights	Not practical — too large for a local workstation
Whisper medium / large-v3	—	Open weights	ASR, not a bounded-task LLM — already in §7.6

Use one model (Llama 3.1 8B or Qwen 2.5 7B) for all three local tasks rather than running several. Hardware sizing, VRAM/RAM requirements, and the GPU-contention question against the DICOM viewer are covered in §8.6.4.

Scope discipline: local hosting only ever covers bounded tasks. Extraction, self-correction, and verification stay on Claude Sonnet regardless of local hardware capability — an 8B open-weight model is measurably weaker at precise instruction-following than a frontier model, and that gap becomes a wrong finding in a signed report, not a rounding error.

When to introduce it: per the earlier volume analysis, local hosting is not worth the setup and maintenance effort below roughly 300–500 reports/day. At pilot volume, run everything on Claude (Sonnet + Haiku split) and defer local hosting to Beta or later.

7.9.5 WHAT GOVERNS ANY FUTURE CHANGE TO THIS SECTION

Every model named here is subject to §8.5’s reliability rule: **exact versions are pinned, never aliased to “latest,” and any change — including a switch prompted by a price drop — is a pipeline version change that must clear the full release gate (§7.7) on the gold set before reaching production.** A cheaper model that has not been measured on your actual dictations is a hypothesis, not a decision.

7.9.6 MODEL CONFIG ADMIN PANEL

A screen for assigning a model to each pipeline task, covering cloud APIs and local open-source models uniformly, with mandatory testing before any assignment can go live. This is the tool §15.0a’s cost-optimization search runs on — the practical mechanism for exercising §7.9.5’s rule.

Layout: one row per task from §7.9.3 (extraction, self-correction, verification, routing pick, classification, shortlist ranking, round-trip check). Each row shows:

- Current model assignment (provider + model name)
- Estimated ₹/report for that task at current volume
- Last gold-set test result (accuracy metrics from §5.2, run against this specific model/task pairing)
- A control to propose a different model

Cloud and local models are the same kind of entry. A model assignment is: provider name, model identifier, an endpoint to call, an auth method, and a cost rate (real per-token pricing for a cloud model, ~₹0 for a local model already running on existing hardware). A local open-source model exposed through an OpenAI-compatible server (the common interface for Ollama, vLLM, and similar local-serving tools) fits the same shape as a cloud provider — only the endpoint and cost rate differ. This means testing “run this task on the free local model instead” is a configuration change, not a code change.

No assignment goes live untested. Proposing a new model for a task triggers an `eval_run` against the frozen gold set (§5.3), scoped to that task, before the assignment can be activated. The panel shows the result — accuracy delta, cost delta — and activation is a separate, explicit step after reviewing it. This is the same release-gate discipline as §7.7 and §8.5, applied at the granularity of a single task rather than the whole pipeline, so a swap on a bounded task (cheap to test, low risk) doesn’t require the same ceremony as a swap on a consequential task (extraction, verification) where a regression matters far more.

Every change is recorded. Which model served which task, from when, and what test result approved the switch — so a later problem can be traced to the exact configuration in force at the time. Data model in §6.14.

Keys never appear in the panel. Provider credentials live in a secrets store; the panel references them by name only.

8. Low-Level Design

8.1 Repository layout

```

radreport/
├── api/                                # FastAPI: ingest, review, admin
│   ├── routes/
│   ├── deps.py
│   └── schemas/                        # request/response models
├── core/
│   ├── config.py                      # pydantic-settings; all model ids pinned here
│   ├── types.py                      # shared domain types
│   └── errors.py
├── pipeline/
│   ├── graph.py                      # LangGraph assembly – the ONLY place edges are defined
│   ├── state.py                      # PipelineState (the graph's shared object)
│   ├── contracts.py                  # Stage protocol
│   └── stages/
│       ├── preprocess.py
│       ├── asr_fanout.py
│       ├── reconcile.py
│       ├── normalise.py
│       ├── segment.py
│       ├── repair.py
│       ├── sketch.py
│       ├── route.py
│       ├── extract.py
│       ├── specialists/
│       │   ├── measurement.py
│       │   ├── laterality.py
│       │   ├── terminology.py
│       │   └── comparison.py
│       ├── ground.py
│       ├── compose.py
│       ├── verify/
│       │   ├── rules.py              # deterministic validators
│       │   ├── critic.py            # LLM critic
│       │   └── roundtrip.py
│       └── confidence.py
├── knowledge/
│   ├── lexicon.py                    # phonetic index, mining, collision audit
│   ├── templates.py                  # registry, versioning, schema compilation
│   ├── retrieval.py                  # hybrid BM25 + vector
│   └── priors.py                     # prior-report lookup
├── adapters/
│   ├── asr/                          # one module per engine, common interface
│   ├── llm/                          # provider abstraction, retry, cost accounting
│   ├── dicom/
│   └── hl7/
├── prompts/                          # versioned, file-based, never inline
│   ├── extract/v3/
│   ├── critic/v2/
│   └── registry.yaml
├── eval/
│   ├── harness.py
│   ├── metrics/                      # one module per §5.2 metric
│   └── gates.py                      # release gate logic
├── db/
│   ├── models.py                    # SQLAlchemy
│   └── migrations/                  # alembic
└── workers/
    └── tasks.py

```

8.2 Core contracts

```
# pipeline/contracts.py
from typing import Protocol, TypeVar, Generic
from pydantic import BaseModel

TIn = TypeVar("TIn", bound=BaseModel)
TOut = TypeVar("TOut", bound=BaseModel)

class StageResult(BaseModel, Generic[TOut]):
    output: TOut
    confidence: float
    cost_usd: float
    duration_ms: int
    model_id: str | None = None
    prompt_version: str | None = None
    warnings: list[str] = []

class Stage(Protocol, Generic[TIn, TOut]):
    name: str
    version: str

    async def run(self, inp: TIn, ctx: "StageContext") -> StageResult[TOut]: ...

    def is_idempotent(self) -> bool: ...
```

Every stage persists a `stage_execution` row via the context. No stage writes domain tables directly; the orchestrator commits. This keeps replay and shadow mode honest.

```
# pipeline/state.py
class PipelineState(BaseModel):
    recording_id: UUID
    pipeline_run_id: UUID
    is_shadow: bool

    audio_meta: AudioMeta | None = None
    asr_runs: list[AsrRunRef] = []
    transcript: Transcript | None = None
    utterances: list[Utterance] = []
    sketch: FindingSketch | None = None
    routing: RoutingDecision | None = None
    field_values: dict[str, FieldValue] = {}
    verification: list[VerificationFinding] = []
    critic_iterations: int = 0 # hard cap enforced in graph.py
```

8.3 Key algorithms

8.3.1 ASR RECONCILIATION

```
INPUT    N ASR hypotheses with word timings + confidences
OUTPUT   reconciled transcript + per-token confidence + disagreement_score
```

1. Align hypotheses into a word transition network (WTN) via iterative pairwise alignment on time-anchored edit distance.
2. For each alignment slot:
 - a. If all engines agree → accept, confidence = mean.
 - b. If majority agrees AND minority confidence < threshold → accept majority (ROVER vote).
 - c. Otherwise mark as DISPUTED.
3. Batch all DISPUTED slots into ONE LLM call with:
 - ±15 words of agreed context around each
 - the candidate variants with per-engine confidence
 - retrieved lexicon candidates matching each variant phonetically
 - the modality (if known) as a prior
 LLM returns a choice per slot plus a confidence, and MAY return "none of these" which escalates the slot.
4. `disagreement_score = disputed_slots / total_slots`

Constraint: the arbitration LLM sees only disputed spans and their local context, never the whole transcript. It selects among candidates; it cannot author new text. This bounds the hallucination surface of the ASR stage.

8.3.2 PHONETIC MATCHING AND CODE-WORD RESOLUTION

```
def resolve_term(surface: str, ctx: MatchContext) -> TermMatch | Escalation:
    # 1. exact surface-variant hit (mined from real transcripts)
    if hit := variant_index.exact(surface):
        return TermMatch(hit.term, 1.0, "exact_variant")

    # 2. phonetic candidates
    keys = double_metaphone(normalise_letters(surface))
    cand_s = phonetic_index.lookup(keys, scope=ctx.active_lexicon_sets)

    # 3. score: phonetic distance + corpus frequency + modality prior
    #           + this speaker's observed usage
    scored = [
        (c, W_PHON * phon_sim(keys, c.phonetic_keys)
         + W_FREQ * c.frequency_score
         + W_MODAL * modality_prior(c, ctx.modality)
         + W_SPKR * speaker_bias(c, ctx.radiologist_id))
        for c in cand_s
    ]
    best, second = top_two(scored)

    # 4. ambiguity guard — a near-tie is NOT a decision
    if best.score < TAU_ACCEPT or (best.score - second.score) < TAU_MARGIN:
        return Escalation(surface, [best, second], reason="ambiguous_match")

    # 5. polysemy deferral: PA / RA / CA resolve only once the
    #       template and section are known
    if best.term.is_ambiguous and ctx.template_version_id is None:
        return Deferred(surface, best.term.candidate_meanings)

    return TermMatch(best.term, best.score, "phonetic")
```

`normalise_letters` handles spelled-out letters ("you gee ess" → UGS , "L M C" → LMC) before phonetic encoding. This is where the E-set confusion between LMC and LMP must be caught — if both are in the lexicon, the margin guard fires and the term escalates rather than silently resolving.

8.3.3 SELF-CORRECTION REPAIR

1. Detect candidate repair sites:
 - lexical markers: sorry, I mean, correction, scratch that, make that, no —, rather
 - structural: immediate restart with high lexical overlap
 - prosodic: pause > 400ms followed by partial repetition
2. For each site, an LLM identifies (reparandum, repair) with the surrounding two utterances as context.
3. Apply: mark reparable utterance superseded_by_id = repair utterance, set is_included_downstream = false on the reparable.
4. NEVER delete. The retracted text stays visible in the review UI as struck-through with the override shown.
5. If the reparable contains a clinical token and confidence in the repair is below threshold → escalate rather than apply.

This exists because “the lesion is in the left — sorry, right kidney” naively filtered leaves `left`. This is a `G4` error class.

8.3.4 TEMPLATE ROUTING CASCADE

```
def route(sketch, study, transcript, recording, radiologist_id) -> RoutingDecision:
    # STAGE 0 – spoken study code: a direct hit, not a filter
    if recording.study_code_template_id and \
        recording.study_code_confidence >= TAU_STUDYCODE:
        chosen = template_repo.get(recording.study_code_template_id)
        # still run coverage validation (Stage 6) – the radiologist can
        # misremember the code just as ASR can mishear it
        if coverage_ok(chosen, sketch):
            return RoutingDecision(chosen, source="study_code", confidence=0.99)
        mark_for_human_template_confirmation("code_content_mismatch")

    # STAGE 1 – hard filter (deterministic)
    if study.study_description:
        # DICOM available
        cans = template_repo.by_study_description(study.study_description)
        source = "dicom"
    else:
        modality = infer_modality_lexical(transcript) # see table below
        region = infer_region_lexical(transcript, sketch)
        cans = template_repo.by_modality_region(modality, region)
        source = "lexical_fingerprint"

    # STAGE 2 – demographic filter (deterministic)
    cans = [c for c in cans
            if study.patient.sex in c.applicable_sex
            and c.age_min <= study.patient.age_years <= c.age_max]

    # STAGE 3 – rank (hybrid retrieval)
    ranked = hybrid_rank(
        query=sketch.as_query(),
        candidates=cans,
        signals={
            "routing_card_similarity": 0.25,
            "exemplar_knn": 0.40, # past dictations – strongest
            "referrer_prior": 0.15, # referring_doctor histogram
            "speaker_prior": 0.10,
            "bm25_terms": 0.10,
        },
    )

    # STAGE 4 – LLM picks from shortlist (never from the full library)
    choice = llm_route(sketch, ranked[:5], study)

    # STAGE 5 – attach finding-triggered modules
    modules = [m for m in module_repo.compatible_with(choice)
               if m.trigger_matches(sketch)]

    # STAGE 6 – coverage validation (the real check)
    orphans = [a for a in sketch.assertions
               if not maps_to_any_field(a, choice, modules)]
    gaps = required_fields_unaddressed(choice, sketch)

    if orphans or choice.confidence < TAU_ROUTE:
        mark_for_human_template_confirmation()

    return RoutingDecision(...)
```

Lexical modality fingerprints (Stage 1 fallback when DICOM is absent):

Modality	High-precision markers
Ultrasound	echogenic, hypoechoic, echotexture, acoustic shadowing, Doppler, transducer
CT	hypodense, hyperdense, Hounsfield, attenuation, arterial/portal/venous phase
MRI	T1, T2, FLAIR, DWI, ADC, signal intensity, hyperintense, gadolinium
Radiograph	lucency, radiopaque, silhouette sign, AP/PA view

These are near-disjoint. A weighted-vote rule over them will be high precision before any model runs.

8.3.5 EXTRACTION WITH PROVENANCE ENFORCEMENT

```
class ExtractedField(BaseModel):
    field_key: str
    value: FieldValue
    assertion_status: Literal["present", "absent", "uncertain", "not_assessed"]
    laterality: Laterality | None
    evidence_char_start: int # REQUIRED – no default
    evidence_char_end: int # REQUIRED
    evidence_quote: str # must appear verbatim in transcript
    confidence: float
```

Post-conditions enforced in `ground.py`, outside the LLM:

```

for f in extracted:
    span = transcript.text[f.evidence_char_start:f.evidence_char_end]
    if normalise_ws(f.evidence_quote) not in normalise_ws(span):
        f.is_grounded = False
        flag(f, "provenance_mismatch")
    if not utterance_at(f.evidence_char_start).is_included_downstream:
        f.is_grounded = False
        flag(f, "sourced_from_excluded_utterance")    # aside / retraction leak

```

Ungrounded fields never render. They appear in the review UI as *proposed but unverified*, which is a different and honest thing.

Self-consistency: run extraction `k=3` (V1) or `k=5` (Beta) at temperature 0.3, vote per field. Disagreement across samples lowers `confidence` and raises `is_flagged`. This is the cheapest use of your latency budget.

8.3.6 ABSENCE HANDLING (THE HIGHEST-LIABILITY LOGIC IN THE SYSTEM)

Two-pass governance (D4). Twenty templates at 15–40 fields is 300–800 `absence_policy` decisions. Presented as one spreadsheet, that yields either a rubber stamp or a stalled project — and a rubber stamp on this specific field is exactly risk R4.

Pass	When	Radiologist effort	Decision
1	Week 5	~30 min	Every field defaults to <code>leave_blank_flag</code> . The radiologist marks only which fields are <code>is_critical</code> (never auto-filled under any circumstance). Short list, fast conversation
2	Post-pilot	~45 min	Query which blank-flagged fields the reviewer filled with identical text >90% of the time. Present ~40 ranked candidates with their observed fill rates . Radiologist promotes selected fields to <code>default_normal</code>

Pass 2 turns a hypothetical decision into an evidence-backed one using the radiologist’s own behaviour, and keeps V1 safe by construction — nothing auto-fills until someone actively promotes it.

```

For each required template_field with no dictated evidence:
    match field.absence_policy:
        "block"                → pipeline flags; field cannot be auto-filled
        "leave_blank_flag"     → value NULL, is_flagged = true           [DEFAULT]
        "default_normal"       → fill default_normal_text,
                               fill_source = 'template_default',
                               ALWAYS visually distinct in review UI

Blanket normals ("rest of the abdomen is unremarkable"):
- only fields with blanket_normal_eligible = true may be filled
- fields with is_critical = true are NEVER blanket-filled
- fill_source = 'blanket_normal', provenance = the blanket utterance
- the blanket phrase is shown alongside every field it filled

```

8.3.7 CONFIDENCE AGGREGATION

```

field_confidence =
    w1 * min(asr_token_confidence over evidence span)
  + w2 * (1 - engine_disagreement over evidence span)
  + w3 * extraction_self_consistency           # agreement across k samples
  + w4 * (1 - normalised_verifier_severity)
  + w5 * speaker_historical_accuracy(field_key, radiologist_id)
  - penalty(fill_source != 'dictated')
  - penalty(is_ambiguous_term_resolved)

report_confidence = min(field_confidence over is_critical fields)
                    * mean(field_confidence over all fields)

```

`min` over critical fields, not mean. One bad critical field must sink the report’s confidence; averaging hides exactly the case that matters.

Thresholds are calibrated on the gold set, per template, and re-calibrated quarterly.

8.3.8 VERIFICATION CHECKS

Deterministic rules (`verify/rules.py`) — cheap, run always:

check_id	Logic	Severity
<code>schema_valid</code>	Payload validates against <code>template_version.json_schema</code>	<code>block</code>
<code>laterality_consistency</code>	Laterality in findings matches impression for the same entity	<code>block</code>

negation_consistency	No entity both present and absent	block
impression_traceable	Every impression item references ≥1 finding field	block
measurement_plausible	Value within per-organ physiological range	warn
unit_normalised	All measurements in template's declared unit	error
sex_anatomy_consistency	No prostate on sex = F, etc.	block
contrast_consistency	Contrast described vs. study.contrast_agent	warn
orphan_assertions	Transcript assertions with no landing field	error
critical_field_grounded	Every is_critical field is is_grounded or explicitly flagged	block
excluded_utterance_leak	No grounded field sources from an aside/retraction	block

LLM critic — tuned for recall, tolerating false positives. The precedent here is a study where GPT-4 detecting clinically significant speech recognition errors in radiology reports achieved an F1 of 86.9% with 76.9% precision and 100% recall, and 94.3% F1 on non-significant errors. Reproduce that operating point: catch everything, accept the review cost, and use `verification_finding.human_verdict` to tighten precision over time.

Round-trip entailment: regenerate a narrative from `structured_payload` alone, then check bidirectional entailment against the report-content utterances. Divergence in either direction is a finding — one direction catches hallucination, the other catches omission.

8.3.9 STUDY-CODE DETECTION

Design the code as a spoken phrase behind a fixed carrier, not as an abbreviation.

```
"Study type: ultrasound abdomen routine."
"Study type: CT chest with contrast."
...
"End of dictation."
```

Rationale, in order of weight:

- 1. **Acoustic evidence.** Spelled-out letters collapse into the confusable E-set (B/C/D/E/G/P/T/V/Z). A code like `UAR` is exactly the failure mode that already threatens `LMC` vs `LMP` — except here an error swaps the entire template. A multi-word phrase carries far more acoustic evidence.
- 2. **Deterministic anchoring.** The carrier `"study type:"` is a fixed segmentation point, so detection is a bounded search in the first ~15 seconds rather than a scan of the whole transcript.
- 3. **Compliance.** Radiologists do not have to memorise a lookup table, and memorisation burden is what actually drives people to skip the habit.

```
def detect_study_code(transcript, window_ms=20_000):
    head = transcript.utterances_within(0, window_ms)
    site = find_carrier(head, CARRIERS)          # "study type", "study is"
    if not site:
        return None                             # non-compliance, not an error
    payload = head.text_after(site, max_words=8)
    cand1 = template_phrase_index.phonetic_rank(payload) # over display phrases
    best, second = top_two(cand1)
    if best.score < TAU_STUDYCODE or (best.score - second.score) < TAU_MARGIN:
        return Escalation(payload, [best, second], "ambiguous_study_code")
    return StudyCode(best.template_id, best.score)
```

`end_marker_detected = false` flags a possibly truncated recording and suppresses post-roll trimming.

Instrument compliance separately from recognition. `CODEWORD_COMPLIANCE` (did they say it) and `STUDYCODE_RECALL` (did we hear it) are different problems with different fixes, and only the first decides D1. Surface per-radiologist compliance to the pilot radiologists — habit formation responds to feedback.

8.3.10 AUTONOMY: GRANT AND REVOCATION

What grouping into classes does, and doesn't, do (D11). This has no effect on the accuracy of any single report — grouping does not make extraction better. What it changes is *how soon* a group of templates can be trusted to skip human review. Proving that an AI report is trustworthy enough to sign without a human check takes roughly 3,000 signed reports' worth of evidence (below). A template doing 5 studies/day would take over a year and a half to accumulate that alone. Ten similar templates pooled at 50/day combined reach the same evidence in about two months. **The benefit is cost-timeline, not quality** — every report is still human-checked until its class earns the right to skip that step.

The sample sizes are larger than intuition suggests. Non-inferiority against a ~2.5% baseline clinically significant error rate, one-sided $\alpha = 0.05$, 80% power:

NI margin	Reports required per class
0.5 pp	~12,000
1.0 pp	~3,000
2.0 pp	~750

Consequence: per-template autonomy is statistically infeasible across most of a 50–200 template library. A template at 1% of volume needs about a year to accrue 750 reports; granting on 40 observations is evidence of nothing. Hence `autonomy_class`.

Only the head of the distribution will ever qualify, and that is an acceptable outcome — the tail is a small share of volume and the highest-variance work. It stays on the human-review path permanently.

Grant — Bayesian sequential, not fixed-n:

```
Model:  theta_ai ~ Beta(alpha0 + cse_count, beta0 + n - cse_count)
Prior:  weakly informative, centred on baseline_cse_rate
Grant   when P(theta_ai < baseline + margin | data) > 0.95
        AND accrued_n >= 0.5 * required_n      # guard against early flukes
Futility when P(...) < 0.20 -> stop accrual, return to human review
Evaluate at every 100th observation.
```

Sequential evaluation lets strong evidence stop early in either direction, which matters when accrual takes months.

Revocation — mechanical, not discretionary:

```
# Bernoulli CUSUM on CSE_SIGNED, evaluated per signed report
p0 = baseline + margin      # acceptable rate
p1 = 2 * p0                 # out-of-control rate
w  = log(p1 / p0)
v  = log((1 - p0) / (1 - p1))

S = max(0, S_prev + (w if is_cse else -v))
if S > cusum_threshold:
    autonomy_class.status = "suspended"      # automatic, no ticket required
    route_all_to_human_review(autonomy_class)
    alert(radiologist_champion, product_owner)
```

Granting autonomy is a statistical decision made by humans. **Revoking it must be a mechanical one made by the system.** A class that starts degrading should return to human review before anyone notices it is degrading.

Operating target: non-inferior at a 1 pp margin against the *measured* transcriptionist baseline, over $\geq 3,000$ reports per class, with continuous CUSUM monitoring after grant.

8.4 Prompt engineering standards

- Prompts live in versioned files under `prompts/`, never inline. `prompt_version` is recorded on every `stage_execution`.
- All extraction prompts use provider JSON-schema mode compiled from `template_version.json_schema`.
- Few-shot exemplars are **retrieved**, not hard-coded: nearest neighbours from `report_draft` history for the same template, same speaker where available.
- Every extraction prompt states the provenance requirement explicitly and includes the transcript with character offsets pre-annotated.
- Negative instruction that matters most: *do not infer, complete, or normalise clinical content that was not spoken; if a field was not addressed, return `not_assessed`.*
- Patient name is never included. `patient.pseudonym`, age, and sex only.

8.5 Reliability

Concern	Mechanism
Idempotency	<code>recording.content_hash</code> unique; <code>asr_run</code> unique on <code>(recording, engine, version, config_hash)</code> ; stages keyed by <code>(pipeline_run_id, stage_name, attempt)</code>
Vendor outage	Queue holds; per-engine circuit breaker; degrade to fewer engines with <code>disagreement_score = null</code> and lowered confidence, never silent single-engine substitution
LLM non-determinism	<code>temperature=0</code> for routing/critic; <code>0.3</code> with k-sampling for extraction; seeds recorded
Silent model updates	Pin exact model version strings. Never use <code>latest</code> aliases. A vendor-side update is a pipeline change and must pass the release gate
Partial failure	Stage-level checkpointing in LangGraph; resume from last successful stage
Cost runaway	Per-run budget cap in <code>pipeline_run.total_cost_usd</code> ; hard abort with alert
Poison message	Max 3 attempts → dead-letter → <code>stat</code> -priority human queue (a failed report still needs a report)

8.6 Training data lifecycle and model adaptation

8.6.1 NOTHING IS TRAINED AT ONBOARDING

The uploaded audio and reports **configure and measure** the system. They do not tune a model. Four reasons, in order of how absolute they are:

Signed reports are not transcripts. This is the blocker, not a preference. A signed report has been reordered, rewoded, cleaned, and de-duplicated. The audio says “*uh, liver is... liver is normal in size, sorry, mildly enlarged*”; the report says “*The liver is mildly enlarged.*” There is no token-level alignment, and token-level alignment is what ASR training requires.

The legacy audio is from the wrong microphones. Adapting to legacy acoustics and deploying on the new standardised push-to-talk hardware trains the model on conditions you are about to stop having. This is the same reasoning as the \$5.3 gold-set split, with a worse consequence: a mismatched gold set misleads you, a mismatched training set degrades you.

It would consume the eval set. At a few hundred pairs, training on them leaves nothing to measure with. The frozen gold set is worth more than the marginal accuracy available from that volume.

The volume is borderline regardless. A few hundred dictations is roughly 10–20 hours — enough to attempt a LoRA, not enough to be confident it beats the base model, and enough to overfit to a handful of speakers.

8.6.2 DATA ASSET INVENTORY

Asset	Volume	Token-aligned to audio	Hardware	Can train
Historical signed reports (unpaired)	thousands	n/a — no audio	n/a	Lexicon, exemplars, routing priors, synthetic transcripts
Legacy paired audio + signed report	few hundred	No	legacy	Nothing directly
Legacy verbatim transcripts (S4)	~100	Yes	legacy	Archive-reprocessing evaluation only
Current-hardware verbatim (gold)	150	Yes	current	Nothing — frozen
Production <code>edit_event</code> with audio spans	grows daily	Yes	current	ASR adaptation — the real corpus
Reviewer utterance corrections	grows	Yes	current	Utterance-classifier distillation
<code>routing_decision.was_human_overridden</code>	grows	n/a	n/a	Router distillation

8.6.3 ADAPTATION ROADMAP

Most of your accuracy in the first six months comes from configuration, not weights.

When	What improves	Mechanism	Data required
Onboarding	Lexicon, templates, ASR biasing, exemplars, routing rules	Configuration + retrieval	Corpus + declared shorthand
Post-pilot, wk 8–12	<code>TAU_*</code> thresholds, confidence weights, keyterm priority	Calibration on dev split	~1,500 edits
Beta, mo 3–6	Utterance classifier, router	Distillation into small supervised models	~5,000 labelled utterances
Beta/GA, mo 6+	ASR	Azure Custom Speech or Whisper LoRA	≥20 h verbatim, current hardware
GA+	Per-speaker ASR	Per-speaker adapters	≥5 h per radiologist

8.6.4 THE VERBATIM COST CURVE

Verbatim transcription is expensive at the start and cheap later, and the reason is compounding:

- **Month 0.** No draft exists. Verbatim means typing from scratch. Expensive, so you buy only the 150 gold items.
- **Month 3.** A decent draft exists. Producing verbatim becomes *verify and lightly correct ASR output* — roughly five to ten times cheaper per minute of audio.
- **Month 6.** `edit_event` rows accumulate automatically as a byproduct of normal review. The corpus builds itself.

The flywheel does not only collect corrections. It makes the expensive input to ASR fine-tuning cheap. That is why month 6 is realistic and month 1 is not.

8.6.5 PREREQUISITE GATES BEFORE ANY ASR ADAPTATION

All six must pass. Recorded in `model_adaptation_run.prerequisite_gates_passed`.

Gate	Condition	Rationale
<code>G1_volume</code>	≥20 h verbatim audio (≥5 h for a per-speaker adapter)	Below this, adaptation is noise
<code>G2_no_leakage</code>	Zero overlap with any frozen <code>eval_set</code>	Enforced by schema <code>CHECK</code> on the snapshot
<code>G3_speaker_balance</code>	≥5 speakers, none exceeding 40% of corpus hours (global adapter only)	Prevents adapting to one voice
<code>G4_hardware_homogeneous</code>	All items share <code>capture_device_class</code> with production	The legacy-audio trap
<code>G5_baseline_measured</code>	Current <code>WER</code> , <code>INS_RATE</code> , <code>CTER</code> recorded on the	You cannot claim improvement without a before

	current gold partition	
G6_legal_basis	Every item has is_training_corpus_eligible = true	DPDP purpose limitation (\$6.11)

8.6.6 EXECUTION

Prefer Azure Custom Speech first. Upload audio plus matching transcripts; training and hosting are handled. Lower effort, no ML engineering pipeline to maintain.

Build the Whisper LoRA pipeline only if Azure underperforms on your gold set. It gives more control and no vendor lock, at the cost of an ML engineer’s sustained attention. Given the team shape in §13, try the managed path first.

Track INS_RATE separately throughout. Adaptation can reduce WER while increasing insertions, which trades a visible error type for an invisible one. In this system that is a regression, not an improvement.

8.6.7 DO NOT FINE-TUNE THE EXTRACTION MODEL BEFORE GA

Extraction is text-to-text, so it needs neither verbatim transcripts nor matching acoustics, and you could train it on synthetic transcripts today. Two reasons not to:

1. At your scale, retrieval-augmented few-shot against a frontier model will likely beat a fine-tuned small model.
2. Templates will change. Retrieval adapts to a new template by indexing new exemplars; a fine-tuned model needs retraining.

Revisit only when template churn has stopped and per-report cost becomes the binding constraint.

8.6.8 RELEASE GATING FOR ADAPTED MODELS

An adapted model is a pipeline change and takes the full \$7.7 path: staging eval gate → shadow for ≥500 reports → canary on one template class → rollout. `relative_improvement ≤ 0` on the gold set is an automatic rejection. Promotion is recorded in `model_adaptation_run` so any report can be traced to the exact model weights that produced it.

9. Gaps — Things Not Yet in Your Plan

Ordered by how expensive they are to retrofit.

9.1 Critical findings communication (regulatory, not optional)

If a dictation contains a critical or unexpected finding (pneumothorax, acute haemorrhage, aortic dissection, free air, new mass), most jurisdictions and accreditation bodies impose a **documented direct-communication obligation** with a time limit. A system that quietly places such a finding in a draft report sitting in a queue can create a compliance failure that did not exist before automation.

Required: - `critical_finding_rule` table: pattern → severity → SLA minutes → escalation path - Detection runs on the transcript, *not* the final report, so it fires before review - `critical_finding_alert` table with acknowledgement tracking and timestamps - Alerts bypass the normal queue entirely

Now scheduled into V1 (week 6). The risk is created the moment reports start queueing, so the mitigation ships with the queue. Rules are radiologist-approved (`critical_finding_rule.approved_by`), not engineer-authored.

9.2 Model version pinning and vendor drift

Cloud model endpoints change beneath you. A silent provider update alters your outputs with no code change and no deploy. Pin exact versions, monitor for deprecation notices, and treat any version change as a release requiring the full eval gate. Without this, your accuracy will drift and you will have no way to attribute the cause.

9.3 Amendments and addenda

Signed reports get corrected. Needs: `final_report.amends_report_id` chain (in the schema above), a distinct amendment workflow, re-export to RIS with the correct HL7 result status, and a policy on whether amendments feed the training flywheel (they should — they are your highest-signal corrections).

9.4 Turnaround SLA and prioritisation

`stat` and `urgent` studies cannot sit behind a routine backlog. `study.priority` is in the schema; the queue must honour it, and SLA breach must alert. Also define behaviour when the pipeline is down: reports must still get produced by the existing manual path.

9.5 Regulatory classification (India)

- **DPDP Act 2023:** patient data and radiologist voice biometrics both in scope. Consent, purpose limitation, breach notification, data principal rights.
- **CDSCO / Software as a Medical Device:** transcription assistance is generally low-risk. **Guideline modules are now out of scope (D6)**, which keeps the product in that band.
- **The boundary to hold in code, not just policy.** Impression is where scope creep enters. An impression that restates dictated

findings is transcription; one that synthesises or infers beyond what was said is decision support. `impression_traceable` is therefore a `block`-severity check: every impression assertion must trace to a dictated finding.

- “Owned by the radiology labs” changes who *operates* the system, not who *built* it. Get the written legal read before Beta regardless.

9.6 Change management

The most common failure mode for this class of system is not technical. It is radiologists refusing to use it after a bad first week, or transcriptionists reasonably fearing for their jobs and disengaging from the review work that the flywheel depends on.

- Be explicit and early with the transcription team about role evolution (reviewing and annotating rather than typing) rather than letting them infer redundancy
- Pilot with two or three willing radiologists, not the whole department
- Publish accuracy numbers internally, including the bad ones
- Give radiologists a one-click “this draft was useless” button and actually track it

9.7 Code-switching

If any radiologist drops into Hindi or a regional language mid-dictation, monolingual English ASR will produce confident nonsense rather than failing loudly. Test for it explicitly. If present, evaluate multilingual or Indic-specialised ASR (AI4Bharat IndicConformer, Sarvam) for those speakers, and add a language-identification pass.

9.8 Data quality gates at ingest

Reject or warn at upload rather than discovering the problem after processing: sample rate below 16 kHz, heavy lossy compression, duration outside expected bounds, SNR below threshold, silence ratio too high, no primary speaker detected. Cheap to build, prevents a category of confusing downstream failures.

9.9 Eval set decay

The frozen gold set ages: new radiologists join, templates change, dictation habits shift. Refresh additively every six months, keeping the original items so trends stay comparable. Track gold-set performance against production edit rates; divergence means the gold set no longer represents reality.

9.10 Also missing

- **Backup and DR** with a tested restore, not just configured backups
- **RBAC** — transcriptionists, radiologists, admins, and auditors need genuinely different access
- **Long-recording handling** — chunking with overlap and cross-chunk context if any dictations exceed the ASR provider’s limit
- **“Report already exists”** — what happens when a recording is uploaded for a study that already has a signed report
- **Cost attribution per template and per radiologist** — you will want it when optimising
- **Onboarding rollback** — if an applied `import_batch` turns out to be wrong (bad template parse, bad merge), there must be a path back. New `template_version` rows rather than mutation makes this possible; the UI for it still has to exist
- **Session-recording safety net** — even though 1 file = 1 report today, add a validator that flags a recording containing two distinct patient contexts
- **Accession number availability** — upload metadata currently carries none. Without it, RIS filing at GA is impossible. Confirm in week 1; this is an integration dependency independent of the DICOM routing question
- **Push-to-talk hardware** — if buying microphones anyway, a dictation device with a record button (Philips SpeechMike class) costs little more than a good headset and solves most of the aside problem in hardware rather than in a classifier

10. Risk Register

#	Risk	Impact	Likelihood	Mitigation
R1	ASR insertion hallucination fabricates a finding	Critical	Medium	VAD gating; <code>INS_RATE</code> tracked separately; multi-engine reconciliation; provenance enforcement; round-trip entailment
R2	Wrong template silently produces a structurally wrong report	High	Medium	Cascade routing; orphan-assertion check; human confirmation below threshold
R3	Self-correction mis-resolved (<code>left</code> retained instead of <code>right</code>)	Critical	Medium	Dedicated repair stage; escalate on low confidence; struck-through display in UI
R4	Auto-filled “unremarkable” on an organ never examined	Critical	Low	Two-pass governance (§8.3.6): nothing auto-fills in V1; <code>is_critical</code> never auto-filled; visual distinction for <code>fill_source !=</code>

				'dictated'
R5	Cross-contamination from a neighbouring radiologist's dictation	Critical	Low	Speaker enrollment + diarization; two-patient-context validator
R6	Vendor model update degrades accuracy silently	High	High	Version pinning; shadow mode; release gates
R7	Transcriptionist disengagement stalls the flywheel	High	Medium	Change management; make review UI fast and rewarding; measure and share time saved
R8	Radiologist rejects the tool after a bad first week	High	Medium	Pilot with volunteers; ship AI-upstream-of-transcriptionist so radiologists see only reviewed output at first
R9	Critical finding delayed in a queue	Critical	Medium	\$9.1 — build detection and alerting in V1
R10	Cost per report exceeds transcription savings	Medium	Low	Per-run budget cap; cost tracked per template; tune k and engine count per class
R11	Gold set too small to detect real regressions	High	Medium	150 items minimum, stratified; supplement with production edit-rate monitoring
R12	Code-word phonetic collision swaps meaning	High	Medium	Scheduled collision audit; margin guard; rename colliding codes
R13	Study-code compliance decays after the novelty wears off	Medium	High	CODEWORD_COMPLIANCE tracked per radiologist and surfaced to them; lexical fallback always runs; DICOM trigger rule (\$4.4) as the backstop
R14	Gold set built on legacy audio misranks ASR engines	High	High if unaddressed	Hardware-split gold set (\$5.3); microphone procurement moved to week 1
R15	Autonomy granted on statistically insufficient evidence	Critical	Medium	Class-level accrual; Bayesian sequential design with a 0.5 x required_n floor; automatic CUSUM revocation
R16	Training corpus retained without a distinct legal basis	High	Medium	is_training_corpus_eligible gate; separate pseudonymised store; explicit disclosure in the privacy notice
R17	Onboarding built as throwaway scripts; unrepeatable when templates change or a second lab arrives	High	High if unaddressed	\$7.8 — versioned, gated, idempotent module from day one
R18	LLM-parsed template schema silently wrong; every report on that template inherits the defect	Critical	Medium	template_import_candidate staging; mandatory radiologist approval; field-by-field review below a parse-confidence threshold
R19	Derived report→template mapping is wrong, corrupting the usage histogram and exemplar index	High	Medium	Hand-verify ≥200 mappings; corpus_template_coverage readiness check at ≥90%
R20	ASR fine-tuned on legacy-microphone audio, degrading production accuracy	High	High if attempted early	Gate G4_hardware_homogeneous ; \$8.6.1 states plainly that onboarding trains nothing
R21	Eval set silently leaks into training, invalidating every downstream number	Critical	Medium	Schema CHECK (excludes_eval_set = true) ; verbatim_transcript.is_eval_set_member permanent exclusion
R22	Adaptation lowers WER while raising INS_RATE , trading visible errors for fabrications	High	Medium	INS_RATE tracked as an independent release-gate metric, not folded into WER

11. Cost Model

Two views: the original per-stage pipeline estimate (kept for the V1/Beta stage breakdown), and the volume-scaled view built from the model comparison in §7.9, which is the number to use for pricing and P&L work.

11.1 Per-stage estimate (component-level)

Per report, assuming ~4 minutes of audio, all-Sonnet baseline before the Haiku split. **Verify all figures against current vendor pricing — these are order-of-magnitude only and this market re-prices within weeks.**

Item	V1 (1 ASR engine, k=3)	Beta/GA (3 ASR engines, k=5)
ASR	~\$0.02	~\$0.05
Sketch + routing	~\$0.02	~\$0.03
Extraction (k samples × sections)	~\$0.15	~\$0.35
Specialists	~\$0.03	~\$0.06
Compose	~\$0.03	~\$0.04
Verify (critic + round-trip)	—	~\$0.10
Storage + compute	~\$0.01	~\$0.02
Total	~\$0.25	~\$0.65

11.2 Volume-scaled cost, final recommended stack

Built from §7.9’s recommended split (Claude Sonnet 5 for consequential tasks, Claude Haiku 4.5 for bounded tasks), the \$7.6 ASR bake-off, and right-sized infra (§ discussion on pilot infra, not Beta/GA infra). Excludes human review cost — see §11.3 for the full picture including it.

Reports/day	Infra ₹/report	ASR ₹/report	LLM (Sonnet + Haiku split) ₹/report	Total ₹/report
20	5.4	3	16	24.4
50	2.2	3	16	21.2
100	2.0	3	16	21.0
200	1.7	3	15	19.7
500	1.7	3	15	19.7
1,000	1.5	3	14	18.5
2,000	1.4	3	13	17.4
5,000	1.2	3	12	16.2

Infra drops fastest early then flattens once past ~100/day. LLM cost drops modestly at volume from provider committed-use pricing. ASR stays flat — it’s a linear per-minute cost with no fixed component to amortize.

With local open-source hosting for bounded tasks (§7.9.4), once volume justifies it (roughly 300-500+/day): the LLM column’s bounded-task share (roughly a third of the ₹12-16) moves to zero marginal cost, run instead on existing radiologist-workstation hardware. Below that volume, the setup and maintenance cost of local hosting exceeds what it saves — stay on Sonnet + Haiku until then.

11.3 With human review — the number that actually matters for pricing

AI-only cost understates what the business needs to charge, because every report still needs proportionate human review (§ pricing discussion). **Transcriptionist review cost: ₹12/report** — a measured figure, replacing the earlier ₹30 (full review) / ₹11 (confidence-weighted blend) estimates.

Reports/day	AI-only ₹/report	+ Transcriptionist review (₹12)	Total ₹/report
20	24.4	12	36.4
100	21.0	12	33.0
500	19.7	12	31.7
1,000	18.5	12	30.5
5,000	16.2	12	28.2

Compare against the actual current cost, now confirmed at ₹12/report (§15.0) — every configuration above this line costs more than that baseline. Pricing therefore cannot be justified by cost savings; see §15.2 for the radiologist-time-saved case, which is now the only remaining path. Track `pipeline_run.total_cost_usd` per template in production so `k`, engine count, and the consequential/bounded split can be tuned per template class rather than globally, using real per-template cost data instead of the flat estimate above — ₹12 is a single blended number, and it may turn out to vary meaningfully by template complexity once measured per-template.

12. First Eight Weeks

Revised for D2 (microphones approved) and D5 (study code adopted). Microphone procurement moves to week 1 because the ASR bake-off must run on the hardware you will actually use.

--	--	--	--

Week	Work	Owner	Output
1	Select and order dictation microphones — prefer push-to-talk models. Blocks the week-3 bake-off	Ops + Eng	Hardware on order
1	Onboarding Console S0-S2 built first — it is the interface for most of weeks 1-5, not a script beside them	Eng	Console skeleton, import + review UI
1	S2: load historical report corpus; derive report→template mapping; usage histogram; radiology provides the pilot 20	Data + Radiology	Ranked list; merge decisions
1	Confirm accession number availability in upload metadata (\$9.10)	Eng + Hospital IT	Go/no-go on RIS filing path
1	Legal read: DPDP, SaMD band, dual legal basis for the training corpus , vendor DPAs	Legal	Written position
1	Audio audit of the legacy archive: sample rate, codec, SNR distribution	Eng	legacy partition scoped
2	Design the study-code phrase set — carrier + spoken template names, phonetically separated. Brief the radiologists	Eng + Radiology	Code phrase sheet
2	Microphones deployed; begin recording the current gold partition	Ops	150 new-hardware recordings underway
2	Brief transcriptionists; begin verbatim transcription (legacy 100 first)	Ops	Gold transcripts
2	S1: import and parse the pilot 20 templates; review dedup proposals; approve schemas	Eng + Radiology	template_version promoted
2	S3: lexicon mining + declared shorthand import; collision audit across shorthand and study codes (LMC / LMP class)	Eng	lexicon_term seeded; blocking collisions resolved
2	Sit with transcriptionists and two radiologists for a day each	Product	Written observations
3	Schema + migrations; ingest API with FLAC enforcement and lossy rejection ; queue	Eng	Deployable skeleton
3	ASR bake-off on the current partition: Deepgram / Azure / Whisper-medium / Whisper-large-v3, ± keyterms. Track WER , INS_RATE , CTER	Eng	Engine decision, evidenced
4	Eval harness + metrics + release gate	Eng	eval_run emitting all \$5.2 metrics
4	S4: paired-audio import; bootstrap ASR run; mine surface variants → re-run S3 (the designed loop)	Eng	lexicon_surface_variant populated
4	Gold annotation complete, both partitions: utterance labels, template IDs, field values	Ops + Radiology	Frozen eval_set
5	Template registry for the pilot 20: schemas, routing cards, spoken study-code phrases	Eng + Radiology	template_version populated
5	absence_policy Pass 1 — 30-minute session: mark is_critical fields only	Radiologist champion	Critical-field list signed off
5	Study-code detection + routing cascade	Eng	ROUTE_TOP1 , STUDYCODE_RECALL measured
6	Extraction with provenance enforcement; grounding + coverage	Eng	FIELD_F1 , HALLUC_RATE measured
6	S5 + S6: mine boilerplate candidates; radiologist authors critical-findings rules, SLAs, escalation paths	Eng + Radiology	boilerplate_candidate ranked; rules approved
6	Critical-findings detection, alerting, acknowledgement	Eng	Alert path live, CF_RECALL measured
7	Review UI: field editing, click-to-listen, flag triage, edit capture, fill_source visual distinction	Eng	Usable by transcriptionists
7	Deterministic validators incl. impression_traceable as block	Eng	Rule suite green
8	Pilot: 2 radiologists, pilot 20, AI draft → transcriptionist → sign	All	EDIT_MINUTES baseline vs. assisted

8	S7: readiness report must pass before pilot go-live	All	Green readiness gate
8	Begin CODEWORD_COMPLIANCE tracking — the four-week clock on D1 starts here	Product	Weekly per-radiologist compliance report

Deliberately not in the first eight weeks: multi-engine ensembling, LLM critic, DICOM, fine-tuning, autonomy. Those are Beta. V1 exists to prove time savings, establish the safety floor, and start the flywheel.

Week 12 checkpoint (D1 resolution): four weeks of pilot compliance data. ≥95% → close DICOM permanently. <90% → schedule for Beta.

13. Team

Role	FTE	Focus
ML / speech engineer	1.0	ASR bake-off, reconciliation, phonetics, fine-tuning
Backend engineer	1.5	Pipeline, data model, integrations
Frontend engineer	1.0	Review UI — do not under-resource this; it decides adoption
Data engineer	0.5	Mining, eval harness, metrics
Radiologist champion	0.2	Template governance, clinical significance grading, absence policies
Transcriptionist lead	0.5	Verbatim annotation, review workflow design
Product / clinical ops	0.5	Change management, pilot coordination

The two part-time clinical roles are the ones most often omitted and most often fatal to omit. Template absence policies and clinical-significance grading cannot be decided by engineers.

14. Decision Log

14.1 Decided

#	Decision	Resolution	Consequence in this document
D1	DICOM integration	Deferred behind a measured trigger	\$4.4 — build V1 without it; CODEWORD_COMPLIANCE decides at week 12. Separate accession dependency raised in §9.10
D2	Microphone standardisation	Approved	Moved to week 1; gold set split by hardware (§5.3); push-to-talk recommended
D3	Pilot templates	Radiology to supply the 20	Week 1 deliverable
D4	absence_policy sign-off	Radiologist	Split into two passes (§8.3.6) so it is 30 min of judgement, not 800 rows
D5	Study-code at start of dictation	Adopted	Designed as a spoken phrase, not an abbreviation (§8.3.9); becomes Stage 0 of routing
D6	Guideline / recommendation modules	Out of scope	Removed from §4.3; impression_traceable raised to block to hold the boundary
D7	Autonomy criteria	Match or surpass human	Formalised as non-inferiority at 1 pp over ≥3,000 reports per class , with automatic CUSUM revocation (§8.3.10)
D8	Audio retention	Store everything	Indefinite lossless FLAC; dual legal basis and separate training corpus (§6.10)

14.2 Resolved (round 2)

#	Decision	Resolution	Consequence
D10	Push-to-talk device or plain headset?	Workflow confirmed as record-then-upload (no live streaming). Physical device (dedicated PTT mic vs. standard headset/laptop mic) still open — default to whatever is in use today, revisit if audio quality in the gold-set bake-off is poor	No blocker; §5.3’s hardware-partitioned gold set still applies whenever the device changes

D11	Which autonomy classes ?	Approach approved — group templates into classes rather than judging each alone (§8.3.10). Actual groupings deferred to Beta, once real usage data exists	No schema change; <code>autonomy_class</code> design unchanged
D12	Measured baseline <code>CSE</code> rate	Audit method specified — see the worked procedure below. Not yet executed	Blocks §8.3.10's non-inferiority math until run
D13	Who grades severity, what sampling rate	Clarified as the same activity as D12 (a person marks each report no-error / minor / significant); frequency to be set once the initial audit is done	No change
D16	Template format	Word and PDF.	Confirms S1 (§7.8.2) parses <code>.docx</code> / <code>.pdf</code> — no RTF or RIS-export parser needed
D17	Does the historical export record which template was used?	No (NA).	Confirms S2's derived <code>corpus_report_template_map</code> (§6.10) is required, not optional — this was already the design's default assumption
D18	Voice enrollment consent	Clarified — a signed form/checkbox before enrollment, since a voiceprint is biometric data under DPDP. Wording still to be drafted	Blocks S0 roster onboarding for any radiologist until signed
D19	Second lab?	Yes, eventually — single hospital now, 100+ tenants later.	Real architecture change. Lightweight <code>tenant_id</code> scoping added now (§6.13); full multi-tenancy (billing, admin, isolation hardening) deferred until a second client signs

D12 worked example — the audit procedure.

1. Pull 50 reports already signed and delivered to patients (unrelated to the pilot — this is about your *current* process, not the AI).
2. For each, a radiologist compares the final report against the original dictation audio and grades it:

Grade	Meaning
No error	Report matches what was said
Minor error	Wrong word, meaning unchanged
Significant error	Wrong side, wrong size, missed/added finding — could change treatment

3. `baseline_cse_rate` = (count of “significant error”) ÷ 50.

This replaces the literature-derived 2.5% estimate used throughout §8.3.10 with your department's actual number, and is the same grading activity D13 asks about — it doesn't need to be a separate exercise once this audit runs.

14.3 Resolved (round 3)

#	Decision	Resolution	Consequence
D9	Accession number linkage	Exists per report; does not chain across a patient's studies.	Prior-study comparison (§4.3 GA scope) marked deferred, build only on customer request — no reliable linkage exists to build it on yet
D11	Does class-grouping improve accuracy?	No. It affects only how soon a group of templates earns the right to skip human review — a cost-timeline lever, not a quality lever	§8.3.10 updated with the explicit clarification, so this isn't re-asked later
D12	Baseline audit — when does it happen?	During client onboarding , folded into S2 of the console (§7.8.2b), gating S7 readiness	No pilot go-live without a measured baseline, per client
D13	Grading owner and sampling rate	Transcriptionist , on a stepped schedule: every report (wks 1-4) → 1-in-5 (wks 5-12) → 1-in-10 (wks 13-16) → 1-in-20 (months 5-7 onward)	§5.4.1 added; <code>edit_event.severity_grade</code> and <code>autonomy_observation.severity_grade</code> ownership corrected throughout the data model
D17	Archive absence, clarified	Confirmed as “no prior version of this software existed” rather than “no historical reports exist” — no design change, closes the earlier NA answer	None
D19	Tenant isolation timing	Real enforcement from day one (application-layer scoping + Postgres RLS), not deferred to a second client. Physical per-tenant data-space separation remains a later, client-managed migration	§6.13 rewritten — isolation is live now, structural split stays future work

14.4 Still open

#	Decision	Needed by	Owner
D14	Critical-findings rule set — which findings, which SLAs, which escalation contacts? (<i>deferred by request — owner to revisit before week 6</i>)	Week 6	Radiologist champion
D15	Transcriptionist role evolution: communicated how, and when? (<i>deferred by request — owner to revisit before week 2</i>)	Week 2	Product + Ops
D21	Draft and finalise voice enrollment consent wording (D18)	Week 2	Legal + Ops
D22	Confirm accession number <i>format/field name</i> in upload metadata so <code>study.accession_number</code> maps correctly (existence is now confirmed by D9; the field mapping itself still needs a concrete check)	Week 1	Eng + Hospital IT
D23	Physical dictation device — dedicated mic or existing headset/laptop? (D10, narrowed)	Non-blocking; revisit if gold-set audio quality is poor	Ops

15. Business Model and Pricing (working notes)

Captured from the economics discussion that followed the technical design. These are working conclusions, not final decisions — flagged separately from §14 because they depend on data not yet collected (market survey, measured radiologist time savings) rather than on engineering choices.

15.0a Project status

Operating as a personal project for now. Offering this commercially to labs/hospitals is contingent on achieving real cost optimization — not a target date. Given §15.0 below, that means: keep building and testing cheaper model configurations, and only revisit the pricing questions in §15.2 onward once a configuration is found that changes the conclusion. The **Model Config Admin Panel** (§7.9.6) exists specifically to make that search fast — swapping models per task and measuring the cost/accuracy result, without a code change each time.

15.0 The cost-savings pitch is retracted — read this before anything else in §15

Every conclusion below §15.1 was originally built on a transcriptionist baseline of ₹40–60/report. **The client has since confirmed the real figure is ₹12/report.** This is not a refinement — it inverts the finding.

	₹/report
Current process (client-confirmed)	12
AI-only, self-hosted split, at 1,000/day (§11.2)	18.5
AI + transcriptionist review, at 1,000/day (§11.3)	30.5

At every configuration in this document, the AI system costs more than the process it would replace. There is no version of “we’re cheaper” available at these numbers. The market-size question in §15.1, the two-baseline framing in §15.2, and the break-even price in §15.4 were all written assuming the opposite was true and need to be read in that light — §15.2’s “cheaper than what you pay now” row no longer exists as an option.

Before treating ₹12 as final, resolve one ambiguity: is it the transcriptionist’s wage cost alone, or the lab’s fully-loaded cost (including overhead, correction cycles, delay cost)? If wage-only, the true comparison point the lab actually feels is higher. If it is genuinely all-in, the conclusion above stands without qualification, and the entire business case now rests on a single unmeasured number — see §15.2a.

15.1 Market size — measure, don’t assume

The known figure (~1,000 reports/day going through transcriptionists) is the size of one city’s current transcriptionist-served volume, not a market ceiling. Two things it doesn’t capture:

- Whether the transcriptionist model is common in other cities — if so, market size scales directly with how many are surveyed
- Labs and radiologists who type their own reports today, with no transcriptionist to displace — a different but real prospect pool

Action: before further build investment, survey 10–15 labs in 2–3 other cities on one question — how do your reports get typed today, and at what cost. The cost half of that question now matters as much as the workflow half, given §15.0 — a ₹12 baseline in one city may not generalise.

15.2 One pricing story remains, and it is now load-bearing rather than optional

Baseline	Status	Pitch	Requires
Transcriptionist cost	Dead, per §15.0	“Cheaper than what you pay now”	—

Radiologist time saved	The only remaining path	“Your radiologist reads more studies per day”	Measured minutes saved per report, with vs. without an AI draft
------------------------	-------------------------	---	---

15.2A WHAT THIS MEASUREMENT NOW NEEDS TO SHOW TO JUSTIFY THE BUSINESS AT ALL

Not just “some time saved” — enough to clear a real bar. Working backward from the AI system’s own cost:

- At ₹30.5/report (AI + transcriptionist review), the radiologist’s saved time needs to be worth **more than ₹18.5/report** just to break even against the ₹12 baseline, before any margin.
- At a loaded radiologist cost of roughly ₹25-50/minute (estimate, unconfirmed — see §15.4), that means **the draft needs to save at least 25-45 seconds per report** merely to reach parity, and meaningfully more to be worth adopting over a process that already works at ₹12.

That’s a far lower bar than the “₹75-150/report” figure floated earlier, which was itself built on the now-obsolete ₹40-60 baseline. The real question the pilot must answer isn’t “does this save time” — it almost certainly does — it’s **“does it save enough time, valued at what a radiologist’s minute is actually worth, to cover an AI system that costs more than doing nothing at all.”**

Action, elevated to the most important open item in this document: measure radiologist minutes saved per report, with vs. without an AI draft, during the V1 pilot (§12, week 8) — and get the loaded radiologist cost figure confirmed rather than estimated (§15.4) before doing the break-even math for real.

15.3 Funding model changes the target economics

Self-funded, selling to labs and hospitals — not venture-funded. This means:

- Engineering cost is founder time, not a headcount line to recover through pricing — the earlier “must clear ₹50L/year team cost” framing does not apply as stated
- At self-funded scale, cutting scope (fewer templates, single ASR engine, no local-model hosting) matters more than optimizing per-report cost, because founder time is the binding constraint, not compute spend
- Ongoing maintenance and support cost (bug fixes, vendor model updates, client support calls) is real and currently unbudgeted in any figure in this document — track it once there are 2-3 paying clients, since support load is the item most often underestimated by a self-funded single-founder operation
- New, given §15.0:** if the radiologist-time-saved case doesn’t clear the bar in §15.2a, self-funding also means there is no team-cost pressure forcing a launch — better to find this out from the pilot than from a client relationship

15.4 Reference numbers from this discussion

All figures are estimates pending real data — treat as planning inputs, not commitments. **Confirmed figures are marked; everything else is still an assumption.**

Item	Value	Status
Current process cost	₹12/report	Confirmed by client — resolve wage-only vs. fully-loaded ambiguity (§15.0)
AI-only cost, self-hosted split, at 1,000/day	~₹18.5/report	Estimate, \$11.2
AI + transcriptionist review, at 1,000/day	~₹30.5/report	Estimate, \$11.3 (₹12 review component confirmed; AI compute component still estimated)
Radiologist loaded cost	₹25-50/minute	Still unconfirmed — estimated from a ₹30-60L/year salary band; get this measured before the §15.2a math is treated as real
Minimum time-saved needed to break even (AI+review vs. current)	~25-45 seconds/report	Derived from the two rows above — recompute once the loaded-cost figure is confirmed

15.5 Open, pending the market survey and pilot measurement

- Whether ₹12 is wage-only or fully loaded (§15.0)** — resolve this first, it changes everything below it
- Radiologist loaded cost per minute — currently an estimate, not measured
- Minutes saved per report with an AI draft — the single most important unmeasured number in this document
- Whether the ₹12 figure holds in other cities, or is specific to this client’s current arrangement
- Final per-report price — now entirely gated on the above, not on beating a transcriptionist cost
- Pricing model shape (per-report vs. subscription vs. per-seat vs. hybrid) — deferred until there’s a viable price to attach it to
- Whether to pursue additional cities before or after the pilot concludes